Deterministic Subset Sum through Signed Circuits and Trace Bounds

icekylinx

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Abstract

We give an exact deterministic algorithm for Subset Sum on arbitrary signed binary integers, with indexed witness recovery. Its running time is $C_{\mathrm{start},3} + \mathrm{poly}(n, L)2^{E_3 n}$, where $E_3 = 158153814247997/319021177919134 < 0.495748$ and the startup constant is finite and independent of the input. Part I proves the signed-circuit compiler and reduction. Part II supplies the preserved two-level theorem F2, with 62 exact contact branches and exponent $E_2 < 0.495836$. Part III adds a third positive-containment lookup and proves the stronger theorem F3, using a separate completed 16-branch exact replay. The combined scalar ledger verifies 112 reflected-profile comparisons and all nine outer costs. Both reconstructed theorems are independent of the older historical contact premise. No simultaneous space exponent is asserted.

Theorem and proof structure

Theorem F3 (third-lookup refinement). On n indexed signed integers of full binary encoding length L, a uniform deterministic algorithm decides feasibility and returns an indexed witness in

$$C_{\mathrm{start},3} + \mathrm{poly}(n, L)2^{E_3 n}, \qquad E_3 = \frac{158153814247997}{319021177919134} < .495748.$$

The startup cost is finite and independent of the input, with no practical numerical bound claimed. All polynomial degrees are fixed and complexity counts deterministic bit operations. Parts I–II prove the inherited signed-circuit, compiler, reduction and witness interfaces. Part III proves the additional lookup, its full-domain acceptance and compiler allowances, and the strengthened complete cost ledger. Its 16-branch receipt records a completed resumable replay under the final source; it is separate from the preserved 62-branch certificate of Theorem F2. The numerical programs discharge their stated finite requirements and do not claim proof-assistant formalization of the manuscript.

Theorem 1 (background conditional composition). Assume the continuous-domain contact inequalities in Part I, Section I.4.2. The older baseline construction decides every signed binary Subset Sum instance and returns an indexed witness in

$$C + \mathrm{poly}(n, L)2^{E_0 n}, \qquad E_0 = \frac{149809813343531}{301754962128739} < 0.4964618056.$$

Here C is the finite input-independent cost of constructing fixed circuit data. The input-dependent term includes residue-cover construction. The imported baseline profile wrapper did not include the modules and receipt needed to replay its full-domain calculation; its reported

leaf counts alone do not establish those contacts. This background theorem retains its explicit premise. Theorems F2 and F3 use their independently reconstructed menus and do not depend on the older numerical receipt. Both compositions use fixed polynomial degrees and the uniform deterministic bit model.

Notation. A witness is a binary vector `x` with `sum_i w_i x_i=t`. The input includes zero, negative and repeated weights. `L` includes every input bit. Logarithms are base two unless `ln` is written. Set `h(z)=-z log_2 z-(1-z) log_2(1-z)` with continuous endpoint values and `H(c,z)=c h(z/c)`.

The finite startup search builds fixed circuit data once; it is part of the running time, with no practical bound claimed here. Original indices remain attached to all representations and are recovered by the signed-circuit witness procedure.

2. Arithmetic primitives

Both algorithms use polynomial-time arithmetic on binary integers, the following prime bound, and exact array operations.

2.1. Prime density and enumeration

For real $x \geq 1$, define

$$\vartheta(x) = \sum_{p \leq x} \ln p, \qquad \psi(x) = \sum_{p^j \leq x} \ln p,$$

where p ranges over primes and j over positive integers. A prime power in $(\lceil n/2 \rceil, n]$ contributes a factor of its underlying prime to $\binom{n}{\lfloor n/2 \rfloor}$. This follows by subtracting the three factorial valuations; its contribution at this power is one and all other contributions are nonnegative. Consequently

$$\psi(n) - \psi(\lceil n/2 \rceil) \leq \ln \binom{n}{\lfloor n/2 \rfloor} \leq n \ln 2.$$

Induction gives $\psi(n) \leq 4n$: for $n \geq 2$, the induction bound is at most $2n + 2 + n \ln 2 \leq 4n$, and the base case is immediate. Thus $\psi(x) \leq 4x$ for real $x \geq 1$.

Conversely, every prime valuation of $\binom{2m}{m}$ is at most the number of powers of that prime not exceeding $2m$. Since the central coefficient is the largest of the $2m+1$ coefficients whose sum is 2^{2m},

$$\psi(2m) \geq \ln \binom{2m}{m} \geq 2m \ln 2 - \ln(2m+1).$$

Put $m = \lfloor x/2 \rfloor$. Integrating the tangent-line bound $1/u \geq 4/3 - 4u/9$ on $[1, 2]$ gives $\ln 2 \geq 2/3$. Thus $\psi(x) \geq x/2$ for $x \geq 64$: it suffices that $\ln(x+1) + 4/3 \leq x/6$, which holds at 64 by $65 < 128$ and $\ln 2 < 1$, and whose right side grows faster thereafter. The prime-power contribution beyond the first power is at most

$$\psi(x) - \vartheta(x) = \sum_{j=2}^{\lfloor \log_2 x \rfloor} \vartheta(x^{1/j}) \leq 4\sqrt{x} \log_2 x.$$

For $x \geq 2^{20}$ the last bound is at most $x/4$ (the ratio $\log_2 x / \sqrt{x}$ decreases in this range). We have proved

$$x/4 \le \vartheta(x) \le 4x \qquad (x \ge 2^{20}).$$

It follows that, for $P \ge 2^{20}$,

$$\#\{p : P < p \le 32P\} \ge \frac{\vartheta(32P) - \vartheta(P)}{\ln(32P)} \ge \frac{4P}{\ln(32P)}.$$

If a nonzero integer has absolute value less than 2^b, at most $b \ln 2 / \ln P$ primes greater than P divide it. The fraction of primes in this interval dividing that integer is therefore at most $2b/P$. This bound holds for all $P \ge 2^{20}$. The interval ratio 32 contributes a fixed factor.

All these primes can be generated deterministically in $P\mathrm{poly}(\log P)$ bit operations. Start with the integers through $\lfloor 32P \rfloor$, and, for each unmarked integer starting at two, mark its proper multiples. A number remaining unmarked is prime: a composite has a smaller prime divisor, already processed. The number of marking iterations is at most $32P \sum_{d \le 32P} 1/d = O(P \log P)$ even if all integers, rather than just primes, are charged. Index arithmetic has $O(\log P)$ bits.

2.2. Exact array and polynomial operations

For two length-q arrays of integers of at most b bits, compute cyclic convolution directly by

$$c_r = \sum_{j=0}^{q-1} a_j b_{r-j \bmod q}.$$

This uses q^2 multiplications and additions on integers of $O(b + \log(q+1))$ bits. The same bound applies to cyclic correlation. Both reductions charge this quadratic cost for their moduli.

In fixed dimension d, a polynomial of coordinatewise degree at most M has at most $(M+1)^d$ coefficients. Store them in an array. Addition, monomial shifts, and multiplication by a fixed polynomial take polynomial time in the degree and coefficient bit length. Integer arithmetic computes binomial coefficients and factorials through order M. These operations support conditional-expectation arguments.

Part I. A faster deterministic algorithm

This part proves the baseline compiler and reduction theorem used by Theorems F2 and F3.

The reduction splits the coordinates into a prefix and two balanced outside parts. A deterministic cover supplies binary–ternary representations of each prefix; exact matching circuits join compatible records at equal integer keys.

I.1. Affine-majorant rebalancing

We develop the colored-circuit framework of Alman and Li [3] and Ye and Li [4], beginning with affine-majorant rebalancing.

I.1.1. A finite rebalancing lemma

Let $\mathcal{I}$ and each $\mathcal{J}_i$ be nonempty finite sets, with positive real pairs

$$(a_{ij}, b_{ij}), \qquad i \in \mathcal{I}, \quad j \in \mathcal{J}_i.$$

Fix real numbers p, q. Suppose that, for every parameter in the closed unit interval,

$$\min_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}_i} a_{ij}^{\tau} b_{ij}^{1-\tau} \le \exp(\tau p + (1-\tau)q), \qquad 0 \le \tau \le 1. \tag{1}$$

For $\lambda > 0$, define

$$F_0(\lambda) = 1 + \lambda,$$

$$F_d(\lambda) = \min_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}_i} a_{ij} F_{d-1}\left(\lambda \frac{b_{ij}}{a_{ij}}\right) \qquad (d \ge 1). \tag{2}$$

The finite minima are attained. Put

$$A_{ij} = a_{ij} e^{-p}, \qquad B_{ij} = b_{ij} e^{-q}, \qquad \delta_{ij} = \ln \frac{B_{ij}}{A_{ij}}, \qquad D = \max_{i,j} |\delta_{ij}|. \tag{3}$$

The constant D is finite.

Lemma I.1.1. Under (1), every integer depth $d \ge 1$ satisfies

$$F_d\Big(e^{(p-q)d}\Big) \le \exp\left(pd + K\ln 2 + \frac{D^2 d}{8K}\right), \qquad K = \lceil \sqrt{d} \rceil. \tag{4}$$

In particular,

$$K\ln 2 + \frac{D^2 d}{8K} \le \left(\ln 2 + \frac{D^2}{8}\right)\sqrt{d} + \ln 2 = o(d). \tag{5}$$

Proof. On nonnegative real functions define

$$(Tf)(t) = \min_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}_i} A_{ij} f(t + \delta_{ij}), \qquad t \in \mathbb{R}. \tag{6}$$

The operator is monotone and positively homogeneous:

$$f \le g \implies Tf \le Tg, \qquad T(cf) = cTf \quad (c \ge 0). \tag{7}$$

The normalized functions

$$G_d(t) = e^{-pd} F_d\Big(e^{(p-q)d+t}\Big) \tag{8}$$

satisfy

$$G_0(t) = 1 + e^t, \qquad G_d = TG_{d-1}. \tag{9}$$

Indeed, substituting (2) into (8) gives (6), because

$$\delta_{ij} = \ln(b_{ij}/a_{ij}) + p - q.$$

For an integer $K \ge 1$, put

$$f_K(t) = \left(1 + e^{t/K}\right)^K, \qquad \phi_K(t) = \ln f_K(t). \tag{10}$$

The binomial theorem gives

$$f_K(t) \geq 1 + e^t. \tag{11}$$

Direct differentiation yields

$$\theta(t) := \phi_K'(t) = \frac{e^{t/K}}{1 + e^{t/K}} \in (0, 1),$$

$$0 \leq \phi_K''(t) = \frac{e^{t/K}}{K(1 + e^{t/K})^2} \leq \frac{1}{4K}. \tag{12}$$

Taylor's theorem gives, for every real δ,

$$\phi_K(t + \delta) - \phi_K(t) \leq \theta(t)\delta + \frac{\delta^2}{8K}. \tag{13}$$

Normalizing (1) gives

$$\min_i \sum_j A_{ij}^{\tau} B_{ij}^{1-\tau} \leq 1. \tag{14}$$

At each t, choose a minimizing list for $\tau = 1 - \theta(t)$. Equations (3), (13), and (14) give

$$\begin{aligned}(Tf_K)(t) &\leq f_K(t) e^{D^2/(8K)} \min_i \sum_j A_{ij} e^{\theta(t)\delta_{ij}} \\ &= f_K(t) e^{D^2/(8K)} \min_i \sum_j A_{ij}^{1-\theta(t)} B_{ij}^{\theta(t)} \\ &\leq e^{D^2/(8K)} f_K(t).\end{aligned} \tag{15}$$

The first inequality holds for each list before minimization, allowing the chosen list to depend on t as in (2).

Keeping K fixed, (7), (9), (11), and (15) imply by induction

$$G_d(t) \leq (T^d f_K)(t) \leq e^{D^2 d/(8K)} f_K(t). \tag{16}$$

Since $f_K(0) = 2^K$, substituting $t = 0$ into (8) and (16) proves (4). The inequalities $\sqrt{d} \leq K < \sqrt{d} + 1$ and $d/K \leq \sqrt{d}$ give (5). □

I.1.2. Tensor circuits

An exact integer depth-two circuit for a finite integer matrix M is a factorization into integer matrices. Its left and right degrees are the maximum nonzero counts in a row of the left factor and a column of the right factor, respectively. Exactness means integer matrix equality, with signed coefficients.

Suppose that every index in a nonempty finite menu supplies a colored circuit for this same matrix. Write its color decomposition as

$$M = \sum_{j \in \mathcal{J}_i} M_{ij}, \qquad M_{ij} = U_{ij} V_{ij}. \tag{17}$$

Remove gates having a zero factor on either side and discard empty colors. Let the color degrees be the positive integers

$$a_{ij} = \deg_L(U_{ij}, V_{ij}), \qquad b_{ij} = \deg_R(U_{ij}, V_{ij}). \tag{18}$$

Proposition I.1.2. For every $d \geq 0$ and $\lambda > 0$, there is an exact integer circuit for $M^{\otimes d}$ satisfying

$$\deg_L + \lambda \deg_R \leq F_d(\lambda), \qquad M^{\otimes 0} = (1). \tag{19}$$

Proof. The one-gate circuit for (1) proves (19) at $d = 0$.

For a positive depth, choose a menu index attaining (2). For each of its colors, apply the induction hypothesis at the smaller depth and the scaled weight

$$\lambda_j = \lambda \frac{b_{ij}}{a_{ij}}.$$

Let the resulting child degrees be L_j, R_j. Tensor that child circuit with the color circuit for M_{ij}, and concatenate the resulting circuits over all colors, using disjoint gate tags. The computed matrix is exactly

$$\sum_j M_{ij} \otimes M^{\otimes(d-1)} = M^{\otimes d}. \tag{20}$$

Tensor products bound endpoint degrees by products, and concatenation by sums; coefficients remain integral. Hence

$$\begin{aligned} \deg_L + \lambda \deg_R &\leq \sum_j (a_{ij} L_j + \lambda b_{ij} R_j) \\ &= \sum_j a_{ij} (L_j + \lambda_j R_j) \\ &\leq \sum_j a_{ij} F_{d-1}(\lambda_j) = F_d(\lambda). \end{aligned} \tag{21}$$

This proves (19). □

Apply the numerical lemma with weight $\lambda = e^{(p-q)d}$. Since the two degrees are nonnegative, (19) bounds them separately:

$$\deg_L \leq \exp\bigl(pd + O(\sqrt{d})\bigr), \qquad \deg_R \leq \exp\bigl(qd + O(\sqrt{d})\bigr). \tag{22}$$

The implicit constants depend only on the fixed menu and affine majorant.

I.2. Biased partial-matching circuits and uniform enumeration

I.2.1. Typed matrices and colored profiles

Fix

$$s = \frac{11}{100}, \qquad t = \frac{9}{20}, \qquad a = \frac{13}{250}, \qquad b = \frac{29}{500}, \qquad a + b = s.$$

A length is admissible if it is a nonnegative multiple of 500. For an admissible length N, let T_N have binary rows of weight tN and ternary columns with aN specified ones, bN specified zeros, and $(1 - s)N$ stars. Its entry is one precisely when the row agrees with every specified

column position. Set $T_0 = (1)$. Every row and column has a matching partner, since $a < t$ and $b < 1 - t$.

For $0 \le x \le c$, write

$$H(c, x) = c \log_2 c - x \log_2 x - (c - x) \log_2 (c - x),$$

with $0 \log_2 0 = 0$. In particular $H(c, x) = ch(x/c)$ when $c > 0$. For a colored exact integer factorization, delete gates with a zero factor and let A_j, B_j be its positive left and right color degrees. Its colored volume is

$$\Upsilon(\tau) = \sum_j A_j^{\tau} B_j^{1-\tau}, \qquad 0 \le \tau \le 1.$$

Section I.1 applies to these exact integer color degrees.

Use the finite lookup menu

$$\rho \in \{13/10, 17/10, 2, 11/5, 12/5, 13/5, 14/5, 31/10, 33/10\}.$$

For each such ρ, put

$$a \le m \le s, \qquad \beta = a + \rho(m - a), \qquad m \le z \le \beta. \tag{23}$$

Throughout this menu, $\beta \le .2434 < t < 1 - t$. The two envelopes are

$$L_\rho(m, z) = \max\{0, H(t, z) + H(1 - t, \beta - z) - H(1 - m, \beta - m)\},$$

$$R_{\text{old}}(m, z) = H(b, m - a) + H(\beta - m, z - m). \tag{24}$$

We form the continuous envelope R_{new} from the two bounds and a simpler certified envelope $\widehat{R}_\rho$.

I.2.2. Exact lookup circuits with simultaneous concentration

The elementary binomial bounds

$$\frac{2^{nh(j/n)}}{n + 1} \le \binom{n}{j} \le 2^{nh(j/n)} \tag{25}$$

follow from the largest term in the binomial distribution of parameter j/n, with the endpoint values interpreted directly.

Fix an integer monomial size $m_0 \in [aN, sN]$, and set

$$B_0 = \lfloor aN + \rho(m_0 - aN) \rfloor, \qquad T_{m_0} = \left\lceil N^2 \binom{N}{m_0} / \binom{B_0}{m_0} \right\rceil.$$

Choose T_{m_0} independent uniform B_0-subsets, retaining an occurrence label when the same subset is chosen more than once. A fixed m_0-set is missed with probability at most e^{-N^2}.

Require two further bounds on the same lists. For every binary row x and integer z_0, the number of occurrences intersecting $\text{supp}(x)$ in exactly z_0 positions is at most $6\nu + 3N$, where

$$\nu = T_{m_0} \frac{\binom{|x|}{z_0}\binom{N-|x|}{B_0-z_0}}{\binom{N}{B_0}}. \tag{26}$$

For every typed query y, with specified sets Y_1, Y_0, and every integer j_0, the number of occurrences satisfying $Y_1 \subseteq B$ and $|B \cap Y_0| = j_0$ is at most $6\nu' + 4N$, where

$$\nu' = T_{m_0} \frac{\binom{bN}{j_0}\binom{(1-s)N}{B_0-aN-j_0}}{\binom{N}{B_0}}. \tag{27}$$

An out-of-domain binomial coefficient is zero. If X is a sum of independent indicators with mean v, then $\mathbb{E}e^X \le e^{(e-1)v}$, and consequently $\Pr(X > 6v + dN) \le e^{-dN} \le 2^{-dN}$. There are at most $2^N \mathrm{poly}(N)$ row events, $3^N \mathrm{poly}(N)$ query events, and $2^N \mathrm{poly}(N)$ coverage events. A union bound proves that all three requirements hold simultaneously at every sufficiently large admissible N. The deterministic compiler below finds such fixed finite circuits by exact search.

Assign every m_0-set U to the first occurrence $B(U)$ containing it. Gates have labels $g = (m_0, B, \sigma)$, where B includes its occurrence label and $\sigma \in \{0,1\}^B$. Define

$$P[x, g] = [x|_B = \sigma],$$

$$Q[g, y] = (-1)^{m_0 - aN} \#\{U : Y_1 \subseteq U \subseteq Y_1 \cup Y_0,\ |U| = m_0,\ B(U) = B,\ \sigma|_U = 1\}. \tag{28}$$

Then $PQ = T_N$ exactly: expanding $\prod_{i \in Y_1} x_i \prod_{i \in Y_0}(1 - x_i)$ by inclusion-exclusion gives precisely (28), with each monomial assigned once. Monomials merged at a fixed gate have the same sign. Their combined coefficient has magnitude at most 2^{bN}.

Color a gate by $(m_0, z_0) = (m_0, |\sigma|)$. Gates with $z_0 < m_0$ have no query neighbor and are removed. There are $O(N^2)$ colors. Each row has at most the count (26) gates of a given color. Counting each contributing monomial separately gives the right-degree bound

$$\binom{bN}{m_0 - aN}\binom{B_0 - m_0}{z_0 - m_0}. \tag{29}$$

Let $m = m_0/N, z = z_0/N, \beta' = B_0/N$. The identity

$$\binom{N}{B_0}\binom{B_0}{m_0} = \binom{N}{m_0}\binom{N - m_0}{B_0 - m_0}$$

and (25) show that the left and right degrees are polynomial times 2^{NL} and $2^{NR_{\mathrm{old}}}$, respectively, with β' substituted for β in (24). The ceiling in T_{m_0} adds at most one to each expected occurrence count and is absorbed in that polynomial.

I.2.3. Counting distinct query gates

Fix a query and a lookup occurrence counted in (27), and put $j = j_0/N$. If its assignment has weight zN and its gate is nonzero at this query, it is one on Y_1 and has at least $(m - a)N$ ones in $B \cap Y_0$. If the latter count is uN, the number of assignments is

$$\binom{jN}{uN}\binom{(\beta - a - j)N}{(z - a - u)N}.$$

Here and in the following calculation β may also denote the actual rounded lookup density β'. Define

$$C(j) = H(b, j) + H(1 - s, \beta - a - j) - H(1 - m, \beta - m),$$

$$S(j, u) = H(j, u) + H(\beta - a - j, z - a - u). \tag{30}$$

Property (27) and (25) bound the number of occurrences by polynomial times $2^{N \max(C(j),0)}$. Therefore

$$R_{\text{agg}}(m, z) = \max_{j,u}\{\max(C(j), 0) + S(j, u)\}, \qquad R_{\text{new}} = \min(R_{\text{old}}, R_{\text{agg}}) \tag{31}$$

is a right-degree exponent. The maximum ranges over the binomial domain and $u \geq m - a$. Each gate assignment is counted once, with harmless overcounting of assignments lacking an assigned monomial. The integer (j, u) pairs add a polynomial factor to the degree bound of the same signed factorization.

For continuity and explicit evaluation, put

$$v = m - a, \quad w = z - a, \quad B = \beta - a, \quad c = 1 - s.$$

We have $0 \leq v \leq b$, $v \leq w \leq B < c$. Let

$$u_1 = \max\{v, bw/(1 - a)\},$$

$$F_1 = H(b, u_1) + H(c, w - u_1) + H(1 - z, \beta - z) - H(1 - m, \beta - m),$$

$$F_2 = \begin{cases} H(B, w), & B \leq b \text{ or } v \leq bw/B, \\ H(b, v) + H(B - b, w - v), & B > b \text{ and } v > bw/B. \end{cases} \tag{32}$$

At $B = 0$, take $F_2 = 0$. Then

$$R_{\text{agg}} = \max(F_1, F_2). \tag{33}$$

Indeed, $\max(\max(C, 0) + S) = \max(\max(C + S), \max S)$. The entropy chain identity and maximization over j give

$$\max_j(C + S) = H(b, u) + H(c, w - u) + H(1 - z, \beta - z) - H(1 - m, \beta - m).$$

For example, after selecting the u ones from the first region and $w - u$ ones from the second, the remaining $B - w$ lookup positions are split optimally between regions of sizes $b - u$ and $c - w + u$; their entropy is $H(b + c - w, B - w) = H(1 - z, \beta - z)$. The unconstrained optimum in u is $bw/(1 - a)$, so imposing $u \geq v$ gives F_1. Its other feasibility constraints hold since $B < c$ and $v \leq \min(b, w)$. For S alone, the unconstrained split in a region of size j has $u = jw/B$. It attains $H(B, w)$ in the first case of (32). In the second case the optimum has $u = v$; the function $H(j, v) + H(B - j, w - v)$ increases up to $j = vB/w > b$, and hence is maximal at $j = b$. The same condition implies $w - v \leq B - b$, so the entropy arguments are valid. All formulas agree at branch boundaries and extend continuously to their compact domain, including $B = 0$.

I.2.4. A rational improvement used in the profile certificate

If $m > a$ and

$$\rho(m-a)^2 \geq b(z-a), \tag{34}$$

then $vB \geq bw$ for the unrounded density $B = \rho v$. Since $w \geq v > 0$, this implies $B \geq b$. Both maxima in (33) occur at $u = v$. The entropy chain identity now gives

$$F_1 = R_{\text{old}} - [H(1-m, z-m) - H(1-s, z-m)],$$

$$F_2 = R_{\text{old}} - [H(\beta-m, z-m) - H(\beta-s, z-m)].$$

For $0 \leq d < x \leq 1$,

$$\frac{\partial H(x,d)}{\partial x} = -\log_2(1-d/x) \geq \frac{d}{x \ln 2} \geq \frac{7}{5}d.$$

The last inequality follows, for example, from $\ln 2 < 5/7$. Integration and continuity at the endpoints prove

$$R_{\text{new}}(m,z) \leq R_{\text{old}}(m,z) - \frac{7}{5}(s-m)(z-m). \tag{35}$$

At $m = a$, condition (34) forces $z = a$, so the zero saving recovers the old bound. On the feasible region, define the saving as follows:

$$\Delta_\rho(m,z) = \begin{cases} \frac{7}{5}(s-m)(z-m), & m > a \text{ and (34) holds,} \\ 0, & \text{otherwise,} \end{cases} \qquad \widehat{R}_\rho = R_{\text{old}} - \Delta_\rho. \tag{36}$$

Thus $R_{\text{new}} \leq \widehat{R}_\rho$. On a rectangular box $[m_0, m_1] \times [z_0, z_1]$ with $m_0 > a$ and $\rho(m_0 - a)^2 \geq b(z_1 - a)$, the certified saving is at least

$$\frac{7}{5}(s-m_1)\max\{0, z_0 - m_1\}. \tag{37}$$

This rational quantity is the only occurrence improvement needed in the finite certificate of Section I.4.

I.2.5. Recursive positive containment and two orientation swaps

Use the direct occurrence circuit on the last profile interval and recursive circuits on the others. We first establish their exact integer matrix identities.

For an outer occurrence B, its local matrix is $I[\sigma, U] = [U \subseteq \sigma]$, where $|\sigma| = zN$ and $|U| = mN$. Complementing both labels and transposing gives positive containment on βN coordinates, with row size $(\beta - m)N$ and query size $(\beta - z)N$. Choose a fixed first inner lookup density γ and put $C = \gamma\beta$. All widths in the profile table below satisfy $\beta - z \leq C \leq \beta$ throughout the outer color domain. Indeed it is sufficient to check $z = m$, and $(\beta - m)/\beta$ increases with m.

Cover every positive query by labelled CN-subsets and assign it uniquely to the first covering occurrence. For a row-intersection color wN,

$$\max\{\beta - z, C - m\} \leq w \leq \min\{\beta - m, C\}. \tag{38}$$

The same sampling and binomial identities as in Section I.2.2 give the uniform positive-containment degree exponents

$$F = \max\{0, H(\beta-m,w)+H(m,C-w)-H(z,C-\beta+z)\}, \quad G = H(C-\beta+z, w-\beta+z). \tag{39}$$

Sample a polynomial times $\binom{\beta N}{(\beta-z)N}/\binom{CN}{(\beta-z)N}$ labelled inner lookups. The expected number with row intersection wN is that sample count times $\binom{(\beta-m)N}{wN}\binom{mN}{(C-w)N}/\binom{\beta N}{CN}$. Its entropy exponent is the expression inside F. A simultaneous Chernoff bound adds a nonnegative floor and a polynomial factor. A positive query assigned to one lookup has exactly $\binom{(C-\beta+z)N}{(w-\beta+z)N}$ matching assignment gates, giving G. Unique query assignment makes the factorization exact.

If $P^{\mathrm{in}}Q^{\mathrm{in}}$ denotes that positive factorization, then

$$I[\sigma, U] = \sum_{h} Q^{\mathrm{in}}[h, B \setminus \sigma] P^{\mathrm{in}}[B \setminus U, h]. \tag{40}$$

For an original signed query (Y_1, Y_0) the coefficient at the new labelled gate (m, B, z, h) is

$$\sum_{\substack{U = Y_1 \cup V,\ V \subseteq Y_0,\ |U| = mN \\ B(U) = B}} (-1)^{|V|} P^{\mathrm{in}}[B \setminus U, h]. \tag{41}$$

Substitution in the inclusion–exclusion identity (28) proves exact matrix equality. The two endpoint exponents are therefore

$$L^{(1)} = L_\rho + G, \qquad R^{(1)} = H(b, m-a) + F. \tag{42}$$

The transpose places G on the outer row side; (42) uses the uncorrected outer occurrence bound.

Positive containment admits the same operation again. In general, on a universe of size n, rows of size Tn and positive queries of size An are covered by lookups of size γn. At row intersection wn the occurrence exponent is

$$L_0 = \max\{0, H(T,w) + H(1-T, \gamma-w) - H(1-A, \gamma-A)\}.$$

The residual matrix, after complement and transpose, has types $T' = (\gamma - A)/\gamma$, $A' = (\gamma - w)/\gamma$. If its degrees per residual coordinate are (L', R'), substitution gives $(L_0 + \gamma R', \gamma L')$. This proves the orientation rule used next.

In units of the original N, put

$$C_1 = \gamma_1 \beta, \quad Q_1 = \beta - z, \quad D_1 = C_1 - Q_1, \qquad \max\{Q_1, C_1 - m\} \le w_1 \le \min\{\beta - m, C_1\},$$

$$F_{1,\mathrm{raw}} = H(\beta - m, w_1) + H(m, C_1 - w_1) - H(z, D_1), \qquad F_1 = \max\{0, F_{1,\mathrm{raw}}\}. \tag{43}$$

For the second inner level choose the fixed rule

$$Q_2 = C_1 - w_1, \quad C_2 = \max\{C_1/2, Q_2\}, \quad D_2 = C_2 - Q_2. \tag{44}$$

When $w_1 < C_1/2$, the choice $C_2 = Q_2$ is exactly the first-level push implementation. Otherwise use the half lookup $C_2 = C_1/2$. This fixed routing has last color and degree contributions

$$\max\{Q_2, C_2 - Q_1\} \le w_2 \le \min\{D_1, C_2\},$$

$$F_2 = \max\{0, H(D_1, w_2) + H(Q_1, C_2 - w_2) - H(w_1, D_2)\}, \qquad G_2 = H(D_2, w_2 - Q_2). \tag{45}$$

Two orientation swaps give

$$L^{(2)} = L_\rho + F_2, \qquad R^{(2)} = H(b, m-a) + F_1 + G_2. \tag{46}$$

At $w_1 = C_1/2$, one has $D_2 = G_2 = 0$ and $F_2 = H(D_1, Q_2)$, so the unmerged formulas agree at the routing boundary. All displayed entropies include their continuous boundary values.

I.2.6. Exact trace aggregation and pooled occurrence counts

Fix an outer color (m, z) and labelled occurrences B and $C \subseteq B$ with $|B| = \beta N$, $|C| = cN$, $c = \gamma_1\beta$. The first inner row is $B \setminus U$, whose intersection with C is $W = C \setminus U$. After the next complement and transpose, the residual query is $K = C \setminus W = U \cap C$. Hence every descendant incidence and its coefficient depends on U only through K for these fixed labels. The coefficient may be summed first:

$$(-1)^{(m-a)N} \#\{U : Y_1 \subseteq U \subseteq Y_1 \cup Y_0, |U| = mN,\ B(U) = B,\ U \cap C = K\}. \tag{47}$$

The aggregated coefficient preserves (41), has $O(N)$ bits, and retains both occurrence labels in its gate.

Write $j = |B \cap Y_0|/N$, $e = |C \cap Y_1|/N$, $f = |C \cap Y_0|/N$, and $q = |K|/N = c - w_1$ in this subsection only. For a fixed pair of occurrences there are at most $\binom{fN}{(q-e)N}$ possible traces. Their full feasibility domain is

$$\begin{gathered} 0 \le j \le b, \quad 0 \le \beta - a - j \le 1 - s, \quad 0 \le e \le a, \quad 0 \le f \le j, \\ 0 \le c - e - f \le \beta - a - j, \quad 0 \le q - e \le f, \quad 0 \le m - a - q + e \le j - f. \end{gathered} \tag{48}$$

The last constraint ensures an outside completion; each feasible trace is counted once regardless of its number of completions.

Choose independent outer samples in polynomial multiplicity times $\binom{N}{mN}/\binom{\beta N}{mN}$, and independently for each labelled outer occurrence choose inner samples in polynomial multiplicity times $\binom{\beta N}{(\beta-z)N}/\binom{cN}{(\beta-z)N}$. Put

$$C_0(j) = H(b, j) + H(1 - s, \beta - a - j) - H(1 - m, \beta - m),$$

$$C_1(j, e, f) = H(a, e) + H(j, f) + H(\beta - a - j, c - e - f) - H(z, c - \beta + z). \tag{49}$$

Here $C_1(j, e, f)$ denotes an exponent, distinct from the lookup width C_1 in (43). The outer histogram has at most a polynomial times $2^{N \max(0, C_0(j))}$ occurrences. Fix any outer family satisfying these bounds, before conditioning on an inner event. Across all eligible labelled outer occurrences, the inner histogram indicators are independent Bernoulli variables with common probability

$$\frac{\binom{aN}{eN}\binom{jN}{fN}\binom{(\beta-a-j)N}{(c-e-f)N}}{\binom{\beta N}{cN}}.$$

A single Chernoff bound on their pooled sum gives at most a polynomial times

$$2^{N\max\{0,C_1(j,e,f),C_0(j)+C_1(j,e,f)\}} \tag{50}$$

pairs. Polynomial oversampling makes every cover, row-degree, and pooled-histogram failure probability $\exp(-\Omega(N^2))$. A union bound over $\exp(O(N))$ queries and labelled families and polynomially many colors proves their simultaneous existence. Independence is used in the unconditioned pooled-histogram estimate.

Consequently the aggregated circuit has unchanged left degree and right exponent at most both the old bound in (46) and

$$G_2 + \max_{j,e,f \text{ in (48)}} [\max\{0, C_1(j,e,f), C_0(j) + C_1(j,e,f)\} + H(f, q-e)]. \tag{51}$$

Both bounds apply to the same integer circuit.

I.2.7. A closed trace correction

Put $d = m - a$, $r = s - m$, $w = w_1$, $q = c - w$ and $D = \beta - m - w$. Suppose

$$\beta \geq s, \quad D \geq 0, \quad 0 \leq q \leq d, \quad w \geq r,$$

$$(b-q)w \geq r(\beta - a - q), \qquad rmD \leq d(m-q)(1-s-D). \tag{52}$$

Then, with $\Delta = H(1-m, D) - H(1-s, D) \geq 0$, the right degree obeys

$$R^{\text{trace}} \leq H(b, m-a) + \max\{0, F_{1,\text{raw}} - \Delta\} + G_2. \tag{53}$$

We prove each branch of (51). Let $k = q - e$. Completion gives $f \leq j - d + k \leq r + k$, so $H(f,k) \leq H(q+r, q) \leq H(b,d)$, proving the zero branch.

For the C_1 branch, increasing j to b only enlarges the finite set of pairs (C, K): relabel neutral positions as prescribed zeros without removing old traces. Put $g = f - k$, so $0 \leq g \leq r$. The expression to maximize becomes

$$H(a, q-k) + H(b,k) + H(b-k, g) + H(\beta - s, w - g) - H(z, c - \beta + z).$$

The unconstrained maximizer in g is $(b-k)w/(\beta - a - k)$. It is nonincreasing in k because $\beta \geq s$; the first inequality on the second line of (52) puts it at least r for $k \leq q$. The constrained maximum is thus at $g = r$. Vandermonde summation over k bounds this branch by

$$B_1 = H(b,d) + H(m,q) + H(\beta - s, D) - H(z, c - \beta + z). \tag{54}$$

For the $C_0 + C_1$ branch, set $u = j - f - (d - k)$. Then $0 \leq u \leq \min(r, D)$. Relax $k \leq d$ to $k \leq d + u$; two Vandermonde summations over k and g give

$$\begin{aligned} J(u) = {} & H(b, d+u) + H(m+u, q) + H(1-s, D-u) \\ & + H(1-m-D, w) - H(1-m, w+D) - H(z, c-\beta+z). \end{aligned} \tag{55}$$

The three terms depending on u are concave, and their derivative at zero is $\log_2[rmD/(d(m-q)(1-s-D))]$. The last inequality of (52) makes it nonpositive. If $r = 0$ or $D = 0$, the feasible interval is already $\{0\}$; otherwise that inequality forces $d > 0$ and $1 - s - D > 0$, while

$m - q \geq a > 0$. This justifies the derivative or covers its boundary exceptions. Thus the maximum is at zero. The entropy chain identity gives

$$B_2 = J(0) = H(b, d) + F_{1,\mathrm{raw}} - \Delta.$$

Finally,

$$B_2 - B_1 = [H(\beta - m, D) - H(\beta - s, D)] - [H(1 - m, D) - H(1 - s, D)] \geq 0.$$

Both brackets integrate $\partial_x H(x, D) = \log_2[x/(x - D)]$ over intervals of length $s - m$, and the first interval lies to the left, where this derivative is larger. The three branch bounds prove (53).

For the rational certificate we use only

$$\Delta \geq \frac{7}{5} \frac{(s - m)D}{1 - m}. \tag{56}$$

Indeed $\log_2[x/(x - D)] \geq D/(x \ln 2) \geq (7/5)D/(1 - m)$ throughout the integral. On a rectangle $m \in [m_0, m_1]$, $v = w_1 - C_1/2 \in [v_0, v_1]$, whose feasible colors satisfy (52), the safe constant subtraction is

$$\delta = \frac{7}{5} \frac{(s - m_1)D_{\min}}{1 - m_0}. \tag{57}$$

The certificate verifies the region conditions throughout each rectangle. Elsewhere set $\delta = 0$ and use (46).

I.2.8. Finite matrices, rounding and the compiler interface

All recursive families have fixed depth and polynomially many colors. At a fixed finite base, exhaustive search finds outer and inner families satisfying the simultaneous finite cover and histogram requirements. Form the coefficients (47), merge equal traces, and store the resulting nonzero finite adjacency lists. Exact multiplication verifies the matrix identity. Each coefficient sums at most $2^{O(N)}$ bounded integer contributions, and consequently has $O(N)$ bits.

Apply the rebalancing and three-count compiler of Sections I.2.10–I.2.16 to these aggregated finite matrices. At the input length, enumerate tensor products of their stored adjacency lists. Inner colors preserve the original row/query counts and each compatible pair's unique joint count description.

For each fixed family the supremum of $\tau L + (1 - \tau)R$ over its colors is convex in τ. Take R to be the minimum of the two valid bounds for that aggregated circuit, with its fixed routing. Convexity gives the contact chords in Section I.4.

Rounding lookup sizes changes normalized parameters by $O(1/N)$. Use the full trace expression (51) and the old bound for this step. Their entropy terms are continuous on compact feasible polyhedra, and limits of rounded feasible colors remain feasible. The maximum over colors therefore has upper error $o(1)$, uniformly in τ. The same conclusion holds at the fixed push/half boundary: the unmerged degree formulas agree there and (51) is a valid bound on either finite construction. Alternatively one can use the old bound in a shrinking neighborhood of that boundary before passing to a sufficiently large base. The color and histogram polynomial factors also contribute $o(1)$ per coordinate.

Section I.2.16 absorbs these fixed polynomial losses into the total compiler allowance $\zeta = 10^{-6}$ by choosing a sufficiently large fixed base. Preprocessing is independent of the Subset Sum instance; variable-length coefficients and gate labels have $O(M)$ bits, with polynomial work per emitted incidence.

I.2.9. Direct profiles, endpoint circuits, and rounding

Define the direct upper lookup profile

$$V_\rho(\tau) = \sup_{(m,z) \text{ in } (23)} \{\tau L_\rho(m,z) + (1-\tau)\widehat{R}_\rho(m,z)\}. \tag{58}$$

It is convex in τ, being a supremum of affine functions. For every fixed $\varepsilon > 0$, the exact colored circuit above, at every sufficiently large admissible N, satisfies uniformly for $0 \le \tau \le 1$

$$\Upsilon_{N,\rho}(\tau) \le 2^{N(V_\rho(\tau)+\varepsilon)}. \tag{59}$$

For rounding, use the continuous actual profile (L, R_{new}). The change in β is below $1/N$, so uniform continuity on the compact domains in (24), (32), and (33) gives a uniform $o(1)$ exponent error. Its unrounded value is bounded by (58); polynomial factors and $O(N^2)$ colors add $O(\log N)/N$.

Two additional exact circuits use one gate for each query or one gate for each row. Their profiles are, respectively,

$$\tau d_{\text{push}}, \qquad (1-\tau)d_{\text{pull}}, \qquad d_{\text{push}} = H(t,a) + H(1-t,b), \quad d_{\text{pull}} = H(1-s, t-a). \tag{60}$$

For the first circuit a row meets $\binom{tN}{aN}\binom{(1-t)N}{bN}$ query gates and each query meets one; for the second, a query meets $\binom{(1-s)N}{(t-a)N}$ row gates and each row meets one. At a fixed base length, the direct and recursive circuits of Sections I.2.2–I.2.8 form an exact menu for T_N. Tensor products and transposes give the mixed menus below.

I.2.10. Constructive simultaneous grouping of three counts

Lemma I.2.1. Given kd integer vectors in $[0,k]^3$, one can deterministically partition them into d groups of exactly k vectors such that every group sum differs in each coordinate by at most $12k$ from k times the global coordinate average. The algorithm takes polynomial time and uses polynomial-bit rational arithmetic in the explicit input size.

Proof. At a node containing qk vectors, split the target number of groups into $q_1 = \lfloor q/2 \rfloor$ and $q_2 = q - q_1$, unless $q = 1$. Start with $x_i = q_1/q$; this solves

$$\sum_i x_i = q_1 k, \qquad \sum_i x_i v_i = (q_1/q)\sum_i v_i, \qquad 0 \le x_i \le 1.$$

While more than four coordinates are fractional, choose five and a nonzero integer null vector for their four constraint columns. Such a vector has $O(\log(k+1))$-bit entries by fixed-dimensional minors (or elimination at the actual rank). Move maximally in one direction until a coordinate reaches zero or one. At most qk moves suffice. Maintaining a common denominator, each move multiplies it by at most one of the bounded null-vector entries, so the total bit length is polynomial. At the end at most four coordinates are fractional, and their sum is an integer. Round exactly this many to one, the others to zero. The cardinality stays exact and each vector sum changes by at most $4k$. Recurse on the selected set and its complement, with deterministic tie breaking throughout.

A node's error relative to its target number of final groups times the global group average becomes $(q_i/q)\Delta + e$ in a child, where $\|e\|_\infty \le 4k$. For every nontrivial split $q_i/q \le 2/3$. Thus a leaf's error is at most $4k\sum_{j\ge 0}(2/3)^j = 12k$. The recursion and all its rational operations have polynomial total size. $\square$

I.2.11. Matching dummy positions and exact joint descriptions

Take k to be a multiple of 500, and put

$$c_{\mathrm{pad}} = 500k, \qquad N_0 = k^2 + c_{\mathrm{pad}}. \tag{61}$$

For a typed row-query pair of length $k^2 d$, divide its coordinates into kd fixed consecutive blocks of length k. Its joint description records in each block the triple (row ones, query ones, query zeros). Apply Lemma I.2.1 to this sequence, with deterministic tie breaking. Each resulting group contains k^2 real coordinates; write its three counts as (U, V, W). They differ from (tk^2, ak^2, bk^2) by at most $12k$ each.

Append c_{pad} dummy positions, with prescribed counts

$$u' = tN_0 - U, \qquad v' = aN_0 - V, \qquad w' = bN_0 - W. \tag{62}$$

These integers satisfy $v', w' \geq 0$, $v' \leq u'$, and $w' \leq c_{\mathrm{pad}} - u'$. Indeed, the four sufficient inequalities are

$$ac_{\mathrm{pad}} \geq 12k, \quad bc_{\mathrm{pad}} \geq 12k, \quad (t-a)c_{\mathrm{pad}} \geq 24k, \quad (1-t-b)c_{\mathrm{pad}} \geq 24k,$$

and all hold for our constants. Put dummy row ones in the first u' positions, query ones in the first v', and query zeros in the last w'. Every dummy position matches. Each padded group therefore belongs to T_{N_0}, and its matching value is exactly the matching value of its real coordinates. The same construction applies to a transposed core by interchanging the endpoint roles.

Joint descriptions preserve exact factorization as follows. Take disjoint circuit copies indexed by all formal descriptions with the required global totals. A row uses only copies with its actual row-count sequence, while a query uses only copies with its actual two query-count sequences. Exactly one description is consistent with both members of any pair. Every other copy has an absent endpoint, so their sum computes the original matrix at every pair.

Among at most $(k+1)^{3kd}$ descriptions, a row enumerates at most $(k+1)^{2kd}$ and a query at most $(k+1)^{kd}$. Their per-coordinate degree overheads are at most $2\log_2(k+1)/k$ and $\log_2(k+1)/k$; use the larger for either mixed endpoint. Padding inflates length by $1 + 500/k$. Both losses vanish as the fixed k increases.

Any remaining admissible length can be padded to the next multiple of k^2 using fewer than k^2 fixed positions of matching type (t, a, b). This is a bounded additive cost once k is fixed; the zero-length matrix is handled separately. These are the restrictions used in mixed-core compilation.

I.2.12. A joint endpoint objective

Let T_N be the typed matrix of Sections I.2.1–I.2.11, and write

$$R = h(t), \quad Q = h(s) + H(s,a), \quad D = R - Q, \quad c = 1 - 2s, \quad S_* = \frac{903}{2500}, \quad C_* = \frac{20}{7}c^2.$$

Section I.4 certifies a piecewise affine function f with the following property: for every fixed $\epsilon > 0$, all sufficiently large admissible N admit a finite menu of exact colored integer circuits for the same matrix T_N, such that the minimum volume at parameter τ is at most $2^{N(f(\tau)+\epsilon)}$. The menu includes the two exact endpoint circuits. Define

$$
\begin{aligned}
R_U(q) &= qR + (1-q)Q, & R_V(q) &= qQ + (1-q)R,\\
v(q) &= \frac{R+Q+S_*}{2} + C_*(q-1/2)^2, & &\\
\delta_U(q) &= v(q) - R_U(q), & \delta_V(q) &= v(q) - R_V(q).
\end{aligned}
\tag{63}
$$

For the baseline parameters, the contact premise of Section I.4.2 and the entropy-compensation argument of Section I.4 give the profile inequality

$$
qf(\tau) + (1-q)f(1-\tau) \le \tau\delta_U(q) + (1-\tau)\delta_V(q) \qquad (0 \le q, \tau \le 1). \tag{64}
$$

Theorem I.2.2 (compiler under the profile hypothesis). Suppose (64) holds. Fix a positive rational allowance η. One terminating deterministic preprocessing program supplies common exact integer adjacency enumerators for

$$
T_{m_1} \otimes T_{m_2}^{\top}
$$

at every pair of lengths divisible by 500. Put $M = m_1 + m_2$ and $q = m_1/M$ when $M > 0$. Their endpoint degrees are bounded by

$$
C_0\, 2^{(\delta_U(q)+\eta)M}, \qquad C_0\, 2^{(\delta_V(q)+\eta)M}, \tag{65}
$$

where C_0 is input-independent. Adjacency lists are generated within these bounds times a polynomial in M. Gate identifiers and integer coefficients have $O(M)$ bits, up to fixed constants. The zero-length matrix is handled separately. The preprocessing depends only on the fixed circuit data and η.

I.2.13. Fixed exact mixed templates

Choose a fixed integer B and a sufficiently large multiple k of 500. Put $N_0 = k^2 + 500k$ and $q_j = j/B$. For every $0 \le j \le B$, consider the fixed matrix

$$
\mathcal{Q}_j = T_{N_0}^{\otimes j} \otimes (T_{N_0}^{\top})^{\otimes(B-j)}.
$$

Tensor independently chosen members of the typed menus. Every resulting menu member computes this same matrix. Transposition exchanges the two degree bounds, so the minimum volume is at most

$$
2^{BN_0[q_j f(\tau) + (1-q_j) f(1-\tau) + \epsilon]}.
$$

By (64) this has the affine majorant with endpoint exponents $\delta_U(q_j) + \epsilon$ and $\delta_V(q_j) + \epsilon$. Apply Lemma I.1.1 and Proposition I.1.2 to these finite color-degree lists. For every sufficiently large depth d, an exact circuit for $\mathcal{Q}_j^{\otimes d}$ has endpoint exponents arbitrarily close to the targets, measured per $BN_0 d$ coordinates. All menu members factor the same exact matrix, with the stated maximum degrees.

There are finitely many grid points. Fix one sufficiently large common depth d_0 and store an exact circuit for each $\mathcal{Q}_j^{\otimes d_0}$. Use these stored circuits thereafter.

I.2.14. Joint descriptions and three-count grouping

Pad each original core to length $k^2 d_i$, with the bounded divisibility conditions specified below. The added lengths are multiples of 500, so fixed matching row/query strings of types t, a, b embed the original matrix as a restriction of the padded matrix.

Apply the grouping construction of Section I.2.11 separately to each core's row-one, query-one, and query-zero counts. It gives d_i groups of k^2 real coordinates, each with discrepancies at most $12k$. Append $500k$ matching dummy positions to each group, yielding T_{N_0} or its transpose.

The padding targets lie within $12k$ of $225k, 26k, 29k$. The integral query-one and query-zero counts therefore fit among the dummy row ones and zeros, respectively.

Tag each restricted grouped circuit by its joint description. The uniqueness argument of Section I.2.11 proves that their sum equals the original matrix.

In the first core, an endpoint on the row side knows its row-count vector and guesses two query-count vectors; an endpoint on the query side does the reverse. The roles are exchanged in the second core. Each endpoint guesses at most $2k(d_1 + d_2)$ counts from $\{0, \dots, k\}$. Hence its logarithmic description overhead is at most

$$\frac{2\log_2(k+1)}{k}\, k^2(d_1 + d_2). \tag{66}$$

The micro-padding multiplies the grouped length by $N_0/k^2 = 1+500/k$. These two overheads tend to zero as the single fixed k increases.

I.2.15. Exact adjacent-template mixing

Choose the initial bounded padding so that

$$d_1 \equiv d_2 \equiv 0 \pmod{d_0}, \qquad d_1 + d_2 \equiv 0 \pmod{Bd_0}.$$

Round each positive core upward to a multiple of $k^2 d_0$, then increase one core by fewer than Bd_0 microgroup units to meet the final condition. Only $O(Bd_0k^2)$ coordinates are added. A zero core can remain zero by rounding in the other core. The case of two zero cores is scalar.

Put $T = (d_1 + d_2)/(Bd_0)$ and $q' = d_1/(d_1 + d_2)$. For $0 < q' < 1$, set $j = \lfloor Bq' \rfloor$ and

$$x = (j+1)T - d_1/d_0, \qquad y = d_1/d_0 - jT.$$

These are nonnegative integers, $x + y = T$, and

$$\begin{aligned} d_1 &= d_0[jx + (j+1)y], \\ d_2 &= d_0[(B-j)x + (B-j-1)y]. \end{aligned}$$

After a permutation, the grouped matrix is exactly

$$(\mathcal{Q}_j^{\otimes d_0})^{\otimes x} \otimes (\mathcal{Q}_{j+1}^{\otimes d_0})^{\otimes y}.$$

At $q' = 0, 1$ use the corresponding endpoint template. Tensoring the stored circuits gives the interpolation of their endpoint exponents. The linear terms of δ_U, δ_V interpolate exactly; their common quadratic term gives, for $\lambda = Bq' - j$,

$$(1-\lambda)\delta_\bullet(q_j) + \lambda\delta_\bullet(q_{j+1}) \le \delta_\bullet(q') + \frac{C_*}{4B^2}. \tag{67}$$

The bounded initial padding changes the logarithmic target by only $O(1)$: the two functions in (63) have bounded derivatives, and both the total length and the first-core length changed by a bounded amount.

I.2.16. Allowances, uniform search, and enumeration

For the stated parameters $C_* = 1521/875$. Take $B = 1000$. Then $C_*/(4B^2) < 4.35 \cdot 10^{-7}$. For $\eta = 10^{-6}$, allocate less than $2.5 \cdot 10^{-7}$ to stored-template errors, less than $2.5 \cdot 10^{-7}$ to (66) and the micro-padding inflation, and less than $5 \cdot 10^{-7}$ to (67), including its micro-padding factor. Choose k sufficiently large, then the common depth d_0. All choices are fixed independently of the input. The same construction works for any prescribed positive rational allowance by first enlarging B.

The grid targets can be tested with rational bounds: writing $x = q - 1/2$ gives $\delta_U = S_*/2 - xD + C_*x^2$ and $\delta_V = S_*/2 + xD + C_*x^2$. Refine a rational interval for D to any fixed tolerance. Dovetail over k, d_0 and tuples of finite integer factorizations, checking exact matrix equality and rational-exponent degree caps by integer-power comparisons. The preceding existence argument proves that this finite preprocessing search terminates.

At the input length, enumerate an endpoint's missing count descriptions, discard invalid ones, and enumerate the corresponding products of stored adjacency lists. The number of descriptions is bounded by (66); valid padded endpoints have positive degree, so enumeration fits (65) up to a polynomial. Tag gates by the description, template choices, and local gate indices. Coefficients are products of fixed stored entries, with $O(M)$-bit lengths.

Compute integer histogram caps by multiplying stored maximum degrees by an integer upper bound on description counts. Their exponents satisfy (65), completing Theorem I.2.2.

I.3. A uniform deterministic reduction

Let $z_1, \dots, z_n, \tau$ be integers of total binary encoding length L. Fix

$$\begin{aligned} s &= \frac{11}{100}, \quad t = \frac{9}{20}, \quad a = \frac{13}{250}, \quad b = \frac{29}{500} = s - a, \\ \mu &= \frac{100000000000000}{301754962128739}, \quad H_0 = 500, \quad \zeta = 10^{-6}. \end{aligned} \tag{68}$$

Write $H(x, y) = xh(y/x)$ and put

$$\begin{aligned} R &= h(t), \quad Q_{\mathrm{ent}} = h(s) + H(s, a), \quad J = 1 - H(s, a) - H(1 - s, t - a), \\ S_* &= \frac{903}{2500}, \qquad v_0 = \frac{R + Q_{\mathrm{ent}} + S_*}{2}, \qquad g_* = \frac{1}{2} + h(s) - J - v_0 - \zeta. \end{aligned} \tag{69}$$

The mixed-core circuit theorem of Section I.2 supplies exact integer factorizations and output-sensitive adjacency enumeration with the following per-input degree bounds. If a fraction q of the main coordinates has the row orientation on side U, define

$$\begin{aligned} R_U(q) &= qR + (1 - q)Q_{\mathrm{ent}}, \qquad R_V(q) = qQ_{\mathrm{ent}} + (1 - q)R, \\ v(q) &= v_0 + \frac{1521}{875}(q - 1/2)^2, \quad \delta_U(q) = v(q) - R_U(q), \quad \delta_V(q) = v(q) - R_V(q). \end{aligned} \tag{70}$$

On M_0 main coordinates the two degree caps are at most a fixed constant times $2^{[\delta_U(q)+\zeta]M_0}$ and $2^{[\delta_V(q)+\zeta]M_0}$, respectively. Enlarge these integer caps to at least one. All circuit constants are input-independent.

Put $M = H_0\lfloor \mu n/H_0 \rfloor$. Handle the fixed finite range of smaller n by exhaustive search, at cost a fixed multiple of $\mathrm{poly}(L)$. Assume henceforth that M meets the stated core ranges.

I.3.1. Three dictionaries and canonical layers

Choose three disjoint M-coordinate blocks. Construct each sorted distinct-sum dictionary by repeated merging of

$$\Sigma(X \cup \{i\}) = \Sigma(X) \cup (\Sigma(X) + z_i),$$

retaining one binary mask per sum. This costs $2^M \mathrm{poly}(n, L)$. Let P be a block with largest dictionary, let $N = |\Sigma(P)|$, and denote the other two blocks by P_U, P_V. For analysis write $N = 2^{(1-d)M}$.

If the exact integer test $N^{50} \le 2^{49M}$ holds, then $d \ge d_0 = 1/50$. Place two complete M-block dictionaries in opposite halves and divide the remaining coordinates equally. Each half has at most $N2^{(n-2M)/2+O(1)}$ entries. Exact complement matching, retaining the stored equal-sum witnesses, decides the instance in

$$\mathrm{poly}(n, L)2^{n/2-dM+O(1)} \le \mathrm{poly}(n, L)2^{(1/2-d_0\mu)n+O(1)}. \tag{71}$$

The unused third block may be split in this branch. Henceforth $d < d_0 = .02$. Let $\mathcal{A} \subseteq \{0,1\}^P$ be the N retained masks of the prefix dictionary; the sum map is injective on $\mathcal{A}$. For a binary prefix c, write $\bar{c} = 1 - c$ and define

$$q_c(K) = \sum_{i \in K} (2c_i - 1)z_i = \sum_{i \in K \triangle \mathrm{supp}(\bar{c})} z_i - \sum_{i \in \mathrm{supp}(\bar{c})} z_i. \tag{72}$$

Let $d_c(k) = |\{b \in \mathcal{A} : |b \oplus \bar{c}| = k\}|$. Assign c to the least layer k with $(M+1)d_c(k) \ge N$, and call its class $\mathcal{C}_k$. The classes partition the binary prefixes. For a nonempty class, put $B = \binom{M}{k}$ and $\kappa = k/M$. Then

$$B \ge \frac{N}{M+1}, \qquad h(\kappa) \ge 1 - d - \frac{\log_2(M+1)}{M}. \tag{73}$$

Since $h(1/5) < .74$, it follows that $1/5 < \kappa < 4/5$ for sufficiently large M.

All classes can be computed in $2^M \mathrm{poly}(M)$ integer operations. For functions on the binary cube define

$$\widehat{f}(S) = \sum_x f(x)(-1)^{S \cdot x}, \qquad (f *_\oplus g)(x) = \sum_y f(y)g(x \oplus y).$$

The identity $\sum_S (-1)^{S \cdot (x \oplus y)} = 2^M 1_{x=y}$ follows by multiplying $1 + (-1)^{x_i + y_i}$ over the coordinates. It proves exact inverse Walsh division by 2^M and $\widehat{f *_\oplus g} = \widehat{f}\widehat{g}$. Splitting on one coordinate computes a transform with $M2^M$ additions. For $A = 1_{\mathcal{A}}$ and $h_k(x) = 1_{|x|=k}$,

$$d_c(k) = (A *_\oplus h_k)(\bar{c}), \qquad \widehat{h}_k(S) = \sum_j (-1)^j \binom{|S|}{j} \binom{M - |S|}{k - j}. \tag{74}$$

A polynomial-size binomial table, one transform of A, and one inverse transform per layer suffice. All intermediate integers have $O(M)$ bits.

I.3.2. Integral cores, label flips, and original integer keys

Fix a nonempty class $\mathcal{C}_k$. Put $M_0 = M - H_0$ and choose

$$m_1 = H_0 \left\lceil \frac{(1-s)M_0 - k}{(1-2s)H_0} \right\rceil, \qquad m_2 = M_0 - m_1, \qquad u_0 = k - sm_1 - (1-s)m_2. \tag{75}$$

Then u_0 is an integer with $0 \le u_0 < (1-2s)H_0 = 390$. Both cores are positive for sufficiently large M, and their lengths are multiples of 500, making all support, row, and query counts integral.

A round consists of a labeled partition $P = H_1 \sqcup H_2 \sqcup R_0$ of sizes m_1, m_2, H_0, together with a flip vector $f \in \{0,1\}^P$. The untyped support family is

$$\mathcal{D} = \{K \subseteq P : |K \cap H_1| = sm_1, \quad |K \cap H_2| = (1-s)m_2, \quad |K \cap R_0| = u_0\}.$$

Its size is

$$K_0 = \binom{m_1}{sm_1}\binom{m_2}{sm_2}\binom{H_0}{u_0} = 2^{h(s)M}\mathrm{poly}(M)^{\pm 1}, \tag{76}$$

uniformly in the layer. This uses $2^{mh(s)}/(m+1) \le \binom{m}{sm} \le 2^{mh(s)}$ and the fixed boundary size. A uniform partition contains each weight-k mask with probability K_0/B: all coordinates, including the boundary, are permuted, and coordinate permutations are transitive on that layer.

For a solution prefix c put $e_i = 2c_i - 1$. On $H_1 \cup R_0$, the U contribution is $e_i 1_{i\in K}$ and the V contribution is $c_i - e_i 1_{i\in K}$. On H_2, the V contribution is $e_i 1_{i\notin K}$ and the U contribution is $c_i - e_i 1_{i\notin K}$. The ternary and binary contributions sum to c_i. Relabel a ternary value $1, -1, 0$ as the query symbol $0, 1, *$, respectively. A ternary value and binary row value sum to a binary value exactly when the row matches its query.

Both core queries have support density s. The original U prefix key is

$$\mathrm{blockkey}_U = \gamma(c, \mathcal{D}) + q_c(K), \qquad \gamma(c, \mathcal{D}) = \sum_{i \in H_2} (1 - c_i) z_i. \tag{77}$$

Indeed on H_2 its contribution is $(1 - c_i) + (2c_i - 1)1_{i\in K}$, and elsewhere the first orientation has no translation. For a fixed c and partition, distinct canonical masks $b = K \oplus \bar{c} \in \mathcal{A}$ give distinct keys in (77), by (72).

A coordinate flip complements its binary row and every specified query bit, preserving the wildcard and compatibility. Retain only flipped rows of weight tm_j and flipped queries with am_j ones and bm_j zeros in core j. The boundary has no type restriction. Undo flips before forming numerical records, so their contributions use the original weights, target, dictionaries, and key identity (77).

For any compatible row/query pair on a core of length m, a uniform flip makes its row uniformly binary. The probability of meeting both type constraints is exactly

$$\theta_m = \frac{\binom{sm}{am}\binom{(1-s)m}{(t-a)m}}{2^m}. \qquad \theta = \theta_{m_1}\theta_{m_2} = 2^{-JM}\mathrm{poly}(M)^{\pm 1}. \tag{78}$$

This probability is independent of the original compatible labels.

The row-oriented proportion on side U is

$$q = \frac{m_2}{M_0}, \qquad \kappa_0 = s + (1-2s)q, \qquad \kappa = \kappa_0 + O(1/M). \tag{79}$$

After reordering coordinates, the two-core matrix is $T_{m_2} \otimes T_{m_1}^{\top}$. Use (70) for this orientation, tensoring with an explicit fixed boundary compatibility matrix. This gives common gate names, exact integer coefficients, and degree caps $D_U, D_V \ge 1$ with

$$\log_2 D_U \le [\delta_U(q)+\zeta]M_0 + O(\log M) + O_{H_0}(1), \quad \log_2 D_V \le [\delta_V(q)+\zeta]M_0 + O(\log M) + O_{H_0}(1). \tag{80}$$

I.3.3. An exact polynomial-tensor round selector

Let $\mathcal{U} \subseteq \mathcal{C}_k$ be the unserved prefixes. For a partition and flip, let W_c count the canonical masks $b \in \mathcal{A}$ such that $K = b \oplus \bar{c}$ belongs to $\mathcal{D}$ and meets both flipped type constraints. Uniform partition and flip averaging gives

$$\mathbb{E}\sum_{c\in\mathcal{U}} W_c \ge |\mathcal{U}|T, \qquad T = \frac{NK_0\theta}{(M+1)B}. \tag{81}$$

A round attaining this mean and all its actual counts can be computed in $2^M \mathrm{poly}(n, L)$ bit operations. To see this, introduce ten formal variables: X_j, Z_j for $j = 0, 1, 2$, recording regional sizes and regional K counts; and Y_j, V_j for $j = 1, 2$, recording core row-one and specified-query-one counts.

For local bits (c_i, b_i) put $k_i = 1 \oplus c_i \oplus b_i$. The binary row before flipping is $1 - b_i$ on core 1 and b_i on core 2. For a local region and flip (j, f_i) use the following monomial as the corresponding entry of a 2 by 2 matrix indexed by (c_i, b_i):

j	entry
1	$X_1 Z_1^{k_i} Y_1^{e_i'} V_1^{k_i e_i'}, \quad e_i' = (1-b_i) \oplus f_i$
2	$X_2 Z_2^{k_i} Y_2^{e_i'} V_2^{(1-k_i)e_i'}, \quad e_i' = b_i \oplus f_i$
0	$X_0 Z_0^{k_i}$.

(82)

At an assigned coordinate retain its chosen matrix; at an unassigned coordinate sum over the six possible region/flip labels. Call the resulting local matrix T_i. The polynomial

$$F = \sum_{c\in\mathcal{U}} \sum_{b\in\mathcal{A}} \prod_{i=1}^{M} T_i(c_i, b_i) \tag{83}$$

counts completions and their canonical witnesses. Extract the coefficient with regional sizes (H_0, m_1, m_2), regional K counts $(u_0, sm_1, (1-s)m_2)$, row-one counts (tm_1, tm_2), and query-one counts (am_1, am_2). The regional K counts already force the global layer.

Compute (83) by applying the tensor product of the local matrices to the length-2^M indicator vector of $\mathcal{A}$, and sum the indicated output entries. Each tensor stage combines pairs of vector entries. Since all exponents are nonnegative, they may be truncated at their target values. There are at most $(M+1)^{10}$ coefficients per polynomial and at most six monomials per local matrix entry. The full tensor computation costs $2^M \mathrm{poly}(M)$ operations. Its coefficients count at most $4^M 6^M$ objects and have $O(M)$ bits.

Expose a coordinate's region and flip simultaneously. If R' coordinates and regional capacities R_0', R_1', R_2' remain, then the number of uniform completions is

$$\frac{R'!}{R_0'!R_1'!R_2'!}\, 2^{R'}. \tag{84}$$

The extracted coefficient divided by (84) is the exact conditional expected witness mass. A feasible child (j, f_i) has probability $R_j'/(2R')$. The parent mean is the weighted average of the child means. Compute them by (83), compare by integer cross multiplication, and choose

a maximizing child. Factorials and comparisons have polynomial bit length. Repeat over all coordinates and extract the terminal target coefficients to obtain all W_c within the claimed cost, proving (81).

There is a uniform upper bound sharper than K_0. In a fixed round, let $c' = c \oplus f$ on one core. On the specified support, the transformed row is $1 - c'$; elsewhere it is c'. A transformed query with am ones and bm zeros therefore chooses am support coordinates among the zeros of c' and bm among its ones. The row-weight condition forces $|c'| = (t - a + b)m$. Put

$$u = t - a + b = \frac{57}{125} = .456, \qquad k_{\text{typed}} = H(1-u, a) + H(u, b).$$

On either core there are at most $\binom{(1-u)m}{am}\binom{um}{bm}$ such supports; on core 2 the support is the complement of K. Including the fixed boundary,

$$W_c \le K_{\text{typed}} \le \text{poly}(M) 2^{k_{\text{typed}} M}, \qquad k_{\text{typed}} < .4981. \tag{85}$$

Use the sharper bound for upper costs, retaining denominator $NK_0\theta$ in (81).

I.3.4. Actual-width buckets, canonical extraction, and implicit energy

Bucket the positive terminal counts into intervals $[w, 2w)$ with w a power of two. Choose a bucket G maximizing $|G|w$. Since there are at most $M + 1$ buckets,

$$|G|w \ge \frac{|\mathcal{U}|T}{2(M+1)}, \qquad w \ge \frac{T}{2(M+1)}. \tag{86}$$

Let K be the exact size of the full typed support family of this round, including the boundary. It is independent of the pattern whenever its transformed regional weights are feasible; patterns with another weight have count zero. By (85), $K \le \text{poly}(M) 2^{DM}$, where

$$D = H(1-u, a) + H(u, b), \qquad \eta = 1 - h(u), \qquad \alpha = 1 + h(s) - s. \tag{87}$$

Expanding the entropy terms of the same four-cell joint distribution in the two orders gives

$$h(u) + D = 1 + h(s) - J, \qquad \frac{K}{T} \le \text{poly}(M) 2^{[d + h(\kappa) - h(u)]M} \le \text{poly}(M) 2^{(d+\eta)M}. \tag{88}$$

Shared canonical extraction. After the round is fixed, the relation between a pattern c and an accepted canonical witness $b \in \mathcal{A}$ is a fixed number of nonnegative coordinate counts,

$$\sum_{i=1}^{M} e_i(c_i, b_i) = \vec{t}, \qquad T_i[c_i, b_i] = X^{e_i(c_i, b_i)}.$$

The ten variables of (82) suffice. For every witness prefix p of length j, apply $T_{j+1} \otimes \cdots \otimes T_M$ to $1_{\mathcal{A}}(p, \cdot)$ and store its polynomial vector F_p. The 2^j vectors each have length 2^{M-j}; all depths cost $2^M \text{poly}(M)$ in time and storage. Degrees are $O(M)$ in a fixed number of variables, and coefficients have $O(M)$ bits. Nonnegative exponents permit truncation above each target.

For fixed c the number of extensions of p is exactly

$$\left[X^{\vec{t} - \sum_{i \le j} e_i(c_i, p_i)} \right] F_p[c_{j+1}, \ldots, c_M], \tag{89}$$

with a negative residual count interpreted as zero. Querying the zero child and comparing its count with a requested lexicographic rank unpacks one witness in polynomial time. Shared $2^M\mathrm{poly}(n,L)$ preprocessing thus reports w witnesses for each of G patterns in $|G|w\mathrm{poly}(n,L)$ additional time. Their integer keys are distinct by canonical injectivity.

One-canonical collision energy. We choose a modulus before reporting witnesses. In a fixed number of regions prescribe the pattern weight U_j and the support counts A_j, B_j on its zero and one positions. Write $\mathcal{R}_c$ for this full typed support family. After fixed complements and orientations, the witness has the form $b(R,c) = R \oplus (1-c)$ and its key is a known translation plus $\sum_i z_i b_i$. Sign changes in z_i and the corresponding permutation of $\mathcal{A}$ handle both orientations. For a positive modulus q define the ordered energy

$$\begin{aligned} E_q(c) &= \sum_{R,R' \in \mathcal{R}_c} 1_{\mathcal{A}}(b(R,c))[d(R,R',c) \neq 0]\,[q \mid d(R,R',c)], \\ d(R,R',c) &= \sum_i z_i (b(R,c)_i - b(R',c)_i). \end{aligned} \tag{90}$$

The orientation change is explicit: on the cores set $c^* = c \oplus f$, $b^* = b \oplus f$ on H_1, and $b^* = b \oplus f \oplus 1$ on H_2. Take $R = K$ on H_1 and $R = \overline{K}$ on H_2. Then $b^* = R \oplus (1 - c^*)$. Replace z_i by $z_i^* = (-1)^{f_i + 1_{i \in H_2}} z_i$ in differences. The affine translation preserves equalities and canonical injectivity. Use this transformed notation below.

Only the first endpoint is canonical. For a bucket with $w \le W_c < 2w$, the aggregate energy at modulus one is less than $2|G|wK$. The oracle below computes every $E_q(c)$ in $2^{\alpha M}\mathrm{poly}(n, L, \log q)$ bit operations, independently of the modulus magnitude apart from $\log q$.

Put $\delta_i = b_i - b_i' \in \{-1, 0, 1\}$. In each region both endpoints have weight $|H_j| - U_j - A_j + B_j$, so their positive and negative difference counts agree. Their support difference has size at most twice $k = \sum_j (A_j + B_j)$, giving at most k negative differences. Fill the ternary array

$$f_q(\delta) = \left[\sum_i z_i \delta_i \neq 0\right] \left[q \mid \sum_i z_i \delta_i\right]$$

on vectors satisfying these budgets, and set it to zero elsewhere. Its input support has at most $\sum_{j \le 2k} \binom{M}{j} 2^j$ entries. Use four count variables x, y, x', y' per region. Direct enumeration of the local bits gives the matrices, with rows c_i and columns b_i,

$$T_0 = \begin{pmatrix} xx' & 1 \\ 1 & yy' \end{pmatrix}, \quad T_+ = \begin{pmatrix} 0 & x' \\ 0 & y \end{pmatrix}, \quad T_- = \begin{pmatrix} x & 0 \\ y' & 0 \end{pmatrix}. \tag{91}$$

The coefficient of $\prod_j x_j^{A_j} y_j^{B_j} (x_j')^{A_j} (y_j')^{B_j}$ in $F_c = \sum_{b,\delta} 1_{\mathcal{A}}(b) f_q(\delta) \prod_i T_{\delta_i}[c_i, b_i]$ is (90) for patterns having the prescribed regional weights U_j; discard the other output entries. Each admissible local pair is counted once.

For a general $T_0 = \left(\begin{smallmatrix} a_0 & b_0 \\ c_0 & d_0 \end{smallmatrix}\right)$, $T_+ = (p_0, q_0)^\top e_1^\top$, and $T_- = (r_0, s_0)^\top e_0^\top$, set

$$\mathcal{D} = a_0 q_0 - c_0 p_0, \quad \mathcal{H} = a_0 d_0 - b_0 c_0, \quad \mathcal{V} = b_0 q_0 - d_0 p_0.$$

The identities $\mathcal{H} p_0 + a_0 \mathcal{V} = b_0 \mathcal{D}$ and $\mathcal{H} q_0 + c_0 \mathcal{V} = d_0 \mathcal{D}$ prove a polynomial rank-three decomposition

$$\mathcal{D}T_\delta[c,b] = \sum_{j=1}^{3} P[j,\delta]U[c,j]V[j,b],$$
$$P = \begin{pmatrix} \mathcal{H} & \mathcal{D} & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \mathcal{D} \end{pmatrix}, \quad U = \begin{pmatrix} p_0 & a_0 & r_0 \\ q_0 & c_0 & s_0 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & 1 \\ \mathcal{D} & \mathcal{V} \\ 1 & 0 \end{pmatrix}. \tag{92}$$

Order the columns of P by $0, +, -$, so channel three receives the negative differences. Transform f_q by $\bigotimes_i P_i$ and the canonical indicator by $\bigotimes_i V_i$, multiply coordinatewise, then contract by $\bigotimes_i U_i$ to obtain $(\prod_i \mathcal{D}_i)F$. Keep strings with at most k channel-three symbols. In the first transform, processed channel-three symbols plus unprocessed negative differences stay within k; in the other transforms, a prefix or suffix exceeding k contributes only to discarded final channels. Every stage therefore has at most

$$R_k = \sum_{j \le k} \binom{M}{j} 2^{M-j}. \tag{93}$$

For example, after t coordinates of the canonical-input transform it is $2^{M-t}\sum_{j\le k}\binom{t}{j}2^{t-j} \le R_k$; the analogous output bound is $2^t\sum_{j\le k}\binom{M-t}{j}2^{M-t-j} \le R_k$. These are ordinary coordinatewise contractions with polynomial arrays. Here $k = sM + O(1)$ and $s < 1/3$, so $R_k \le \mathrm{poly}(M)2^{[1+h(s)-s]M}$. The initial difference table is smaller because $h(2s) + 2s < 1 < \alpha$.

For (91), $\mathcal{D} = x'(xy - 1)$ is a nonzero formal polynomial. The common multiplier is $\prod_j (x'_j)^{m_j}(x_jy_j - 1)^{m_j}$. Remove the monomial factors and multiply by the truncated series

$$(1 - x_jy_j)^{-m_j} = \sum_{\ell \ge 0} \binom{m_j + \ell - 1}{\ell} (x_jy_j)^\ell \tag{94}$$

with the appropriate sign. Known divisibility gives the exact F_c. The computation uses formal polynomials of degree $O(M)$ in a fixed number of variables, with polynomial coefficient lengths.

Enumerate the pattern and two support masks on the fixed H_0-boundary. Restrict the canonical indicator to the first boundary witness, add the exact boundary difference to the test for f_q, and sum the two-core transforms. These constantly many computations preserve the core budgets, state exponent, and complete zero-difference exclusion. The witness family uses the original cores and boundary, proving the signed-key oracle.

Fixed-factor modulus. Choose a polynomial bit bound $B_0 \ge 1$ for all nonzero complete key differences. If $K = 1$, take $Q = 1$. Otherwise fix $r = 64$, and set

$$t_0 = \lceil (4K)^{1/r} \rceil, \qquad P_0 = 2^{32}(B_0 + r + 2)^2 t_0. \tag{95}$$

Sieve the primes in $(P_0, 32P_0]$. Section 2.1 gives at least $P_0/\log_2 P_0$ candidates, more than $2r$. A nonzero B_0-bit difference has at most $B_0/\log_2 P_0$ prime factors in the pool. At least half the pool remains unused at every stage; therefore at most a $2B_0/P_0 \le 1/(2t_0)$ fraction of the unused primes divide it. Starting at $m = 1$, compute $\sum_{c\in G} E_{mp}(c)$ for each unused candidate and choose a minimum. Conditional averaging reduces the surviving aggregate by a factor at least $2t_0$ at each stage. The product Q of the r chosen distinct primes satisfies

$$K \le Q \le K\mathrm{poly}(n, L), \qquad \sum_{c\in G} E_Q(c) \le \frac{2|G|wK}{(2t_0)^r} \le |G|w/2. \tag{96}$$

Indeed $Q > P_0^r \geq t_0^r \geq 4K$ and $t_0 \leq 2(4K)^{1/r}$; r is fixed, so all factors other than K in $(32P_0)^r$ are polynomial. The number of candidate oracle calls is at most $K^{1/64}\mathrm{poly}(n, L)$. Their total time per round is

$$2^{(\alpha+D/64)M}\mathrm{poly}(n, L). \tag{97}$$

Prime sieving and the scan of each output energy vector are smaller. The conditional prime argument is an instance of the implicit-counting multi-prime method of Jin, Park, Saha, and Xu [9]; the oracle (90)–(94) and its costs are supplied here.

Serve the explicit patterns with $E_Q(c) \leq 2w$; at least half of G qualify. Extract w canonical witnesses per served pattern by (89). Their ordered modular collisions are a subset of (90). If n_y are their residue multiplicities, then

$$\sum_y n_y = w, \qquad \sum_y n_y^2 \leq w + E_Q(c) \leq 3w, \qquad |S_c| \geq w/3 \geq w/5, \tag{98}$$

where S_c is the deduplicated residue support. Cauchy–Schwarz gives this bound; use $w/5$ in the cover below.

I.3.5. Complete local covers and harmonic charging

For this fixed round initialize $U_c = \mathbb{Z}_Q$ for every served pattern. A residue δ newly covers

$$A_f(\delta) = \sum_c \sum_{s \in S_c} 1_{U_c}(\delta - s), \qquad \frac{1}{Q}\sum_\delta A_f(\delta) = \sum_c \frac{|U_c||S_c|}{Q} \geq \frac{w}{5Q}\sum_c |U_c|. \tag{99}$$

Store uncovered indicators in binary range-sum trees. On a candidate interval I, sum the counts on $I - s \pmod Q$ over the stored S_c; each shifted interval splits into at most two ordinary intervals. Bisect I, compare child means by cross multiplication, and retain a child with at least the parent mean. After $O(\log(Q+1))$ steps, including odd interval lengths, a singleton attains the original mean. Delete its covered points and repeat.

If H patterns are served, at most

$$D_i \leq \left\lceil \frac{5Q_i}{w_i}\left(1 + \lceil \log_2(HQ_i) \rceil\right) \right\rceil \tag{100}$$

choices suffice. Indeed each step multiplies the uncovered count by at most $1 - w_i/(5Q_i)$; using $1 - x \leq e^{-x}$ makes the final integer count less than one. Initialization costs $|G_i|Q_i$, and each search and deletion pass costs $|G_i|w_i\mathrm{poly}(n, L)$. Thus all local covering work and state cost $|G_i|Q_i\mathrm{poly}(n, L)$, including the residue deduplication. For $w_i = Q_i = 1$, use residue zero. In general,

$$D_i \leq \mathrm{poly}(n, L)K/w_i, \qquad \frac{D_i}{Q_i} \leq \frac{\mathrm{poly}(n, L)}{w_i}. \tag{101}$$

Remove the served patterns and repeat the selector and bucket construction. Write $C_i = |\mathcal{U}_i|$, let G_i be the selected bucket, and let H_i be its served subset, so $|H_i| \geq |G_i|/2$ and $C_{i+1} = C_i - |H_i|$. Equation (86) implies

$$\frac{1}{w_i} \leq \frac{4(M+1)}{T}\frac{|H_i|}{C_i}.$$

For every nonfinal iteration $|H_i|/C_i \leq \int_{C_{i+1}}^{C_i} dx/x$, and the final term is at most one. Since the served sets are disjoint,

$$\sum_i |G_i| \le 2|\mathcal{C}_k|,$$
$$\sum_i \frac{1}{w_i} \le \frac{4(M+1)(1+\ln|\mathcal{C}_k|)}{T} \le \mathrm{poly}(M)\frac{B}{NK_0\theta}, \tag{102}$$
$$I \le K\sum_i \frac{1}{w_i} \le \mathrm{poly}(M)K/T \le \mathrm{poly}(M)2^{(d+\eta)M}.$$

Combining (101) and (102) gives the full filtering rate

$$\boxed{\sum_i \frac{D_i}{Q_i} \le \mathrm{poly}(n,L)\frac{B}{NK_0\theta}.} \tag{103}$$

Every local residue repetition is included. Equations (86), (96), and (88) also give

$$\max_i D_i \le \mathrm{poly}(n,L)K/T \le \mathrm{poly}(n,L)2^{(d+\eta)M}.$$

The total reported-witness and complete-cover costs are at most

$$\sum_i |G_i|w_i + \sum_i |G_i|Q_i \le 2^M K\mathrm{poly}(n,L). \tag{104}$$

Shared witness-table preprocessing and selectors cost $I2^M\mathrm{poly}(n,L)$. All candidate-prime oracle calls cost $I2^{(\alpha+D/64)M}\mathrm{poly}(n,L)$. These charges include the full-family modulus scale and its residue states, in addition to harmonic filtering.

Each assigned round covers every residue for every served prefix. A common shift of its local residue sequence preserves coverage by translating the residue to be covered oppositely. Choose the modulus and shift separately in each round.

I.3.6. Cached outer lists and balanced incidence bounds

Split the coordinates outside P into O_U, O_V of sizes differing by at most one, containing the whole blocks P_U, P_V, respectively. This is feasible since $3M \le n$. Cache the two exact lists using each designated dictionary and all binary masks on the other coordinates of that side. Retain the corresponding witnesses. Their sizes are at most

$$|\mathcal{O}_U|, |\mathcal{O}_V| \le 2^{(n-M)/2-dM+O(1)}. \tag{105}$$

For a selected round, let $\mathcal{B}_U, \mathcal{B}_V$ be its allowed typed prefix records, after undoing its flips. Their exact full keys are

$$\mathrm{key}(u) = \mathrm{blockkey}(u) + \mathrm{outersum}(u), \qquad \mathrm{key}'(v) = \tau - \mathrm{blockkey}(v) - \mathrm{outersum}(v). \tag{106}$$

The binary row universe has size $\binom{m}{tm} \le 2^{Rm}$ and the query universe has size $\binom{m}{sm}\binom{sm}{am} \le 2^{Q_{\mathrm{ent}}m}$. Together with (80), each weighted prefix universe obeys

$$D_U|\mathcal{B}_U|,\ D_V|\mathcal{B}_V| \le \mathrm{poly}(M)2^{[v(q)+\zeta]M_0+O_{H_0}(1)}. \tag{107}$$

Since $h''(x) = -1/[\ln(2)x(1-x)]$ and $h'(1/2) = 0$,

$$1 - h(1/2+x) \ge \frac{2}{\ln 2}x^2 \ge \frac{20}{7}x^2.$$

The last inequality uses $\ln 2 < 7/10$. Because $(20/7)(1-2s)^2 = 1521/875$, (79) gives

$$\frac{1521}{875}(q-1/2)^2 \le 1 - h(\kappa_0).$$

The relevant layers stay in $(1/5, 4/5)$ and $\kappa - \kappa_0 = O(1/M)$, so $M|h(\kappa) - h(\kappa_0)| = O(1)$. Thus

$$D_U|\mathcal{B}_U|,\ D_V|\mathcal{B}_V| \le \mathrm{poly}(M)2^{[v_0+1-h(\kappa)+\zeta]M}. \tag{108}$$

For modulus Q_i, rebuild the cached outside residue buckets and histograms. Enumerate the raw prefix patterns to form prefix-key histograms, convolve them directly with the outside histograms, and reflect the V sums around τ. The resulting nonnegative weighted-load array is

$$W_i(r) = D_U\#\{u : \mathrm{key}(u) \equiv r\} + D_V\#\{v : \mathrm{key}'(v) \equiv r\}.$$

For the local cover $\delta_{i,1}, \dots, \delta_{i,D_i}$, scan all common shifts and choose σ_i minimizing $\sum_j W_i(\delta_{i,j}+\sigma_i)$. Its value is at most

$$\frac{D_i}{Q_i}\left(D_U|\mathcal{B}_U||\mathcal{O}_U| + D_V|\mathcal{B}_V||\mathcal{O}_V|\right). \tag{109}$$

Direct cyclic convolution and shift evaluation cost $O(Q_i^2)$ and $O(Q_iD_i) \le Q_i^2\mathrm{poly}(n, L)$ integer operations, respectively.

For each chosen residue and each U prefix key x, access only the outside bucket at $\delta_{i,j}+\sigma_i-x$. On side V access the bucket at $\tau - x - \delta_{i,j} - \sigma_i$. Emit its entries and only then enumerate each retained record's circuit adjacency. Raw prefix enumeration and bucket lookups pay the overhead; (109) pays every emitted record because both degree caps are at least one.

Sum (109) and use (76), (78), (103), (105), (108), and $B \le 2^{Mh(\kappa)}$. The logarithm of the retained incidence volume, up to polynomial factors, is at most

$$\begin{aligned}(n-M)/2 - dM + [v_0 + 1 - h(\kappa) + \zeta]M \\ + [h(\kappa) - (1-d) - h(s) + J]M = n/2 - g_*M.\end{aligned} \tag{110}$$

Both the deficiency and the layer entropy cancel. The outside lists remain balanced for every layer.

I.3.7. Exact counting and binary witnesses

For each selected round and each local residue occurrence, use the common exact integer compatibility circuit on the retained records. Gate tags include both that round and that residue occurrence, or these instances are processed separately. Sort incidences by their full integer key and gate, aggregate the coefficients on each side, and sum the products of the two aggregates. Circuit exactness gives the nonnegative integer

$$\sum_{u,v} 1_{\mathrm{key}(u)=\mathrm{key}'(v)} 1_{\text{coordinatewise compatible}(u,v)}. \tag{111}$$

Accept when the completed count is positive. A compatible equal-key pair reconstructs a binary prefix and two stored outside witnesses with total sum τ.

Conversely, fix a genuine solution with prefix c. Its assigned good layer eventually serves it. The selected canonical keys are actual representations meeting the typed domains, and remain distinct by (72), (77). Its assigned complete local cover applies to every outside residue, also

after the common shift. Consequently a compatible representation has its U key in a selected residue. The target-adjusted V key is exactly equal. Replacing either designated outside block by its stored equal-sum witness preserves both keys. Undoing the flips before forming the keys preserves the original input sums. Hence this local instance has positive count. If every completed count is zero, no solution exists.

Recover a pair from a positive count by bisecting the retained U list and recounting, keeping a positive part, until one record remains; then bisect the V list. Restricted counts retain the nonnegative interpretation and sum to the preceding count. The polynomially many restrictions each reduce incidence volume, so sorting, aggregation, and recovery add polynomial factors to (110).

I.3.8. All remaining costs and uniformity

The remaining operations have the following bounds up to $\mathrm{poly}(n, L)$ factors. Here $I \le 2^{(d+\eta)M}\mathrm{poly}(n, L)$, $Q_i \le 2^{DM}\mathrm{poly}(n, L)$, and $\max_i D_i \le 2^{(d+\eta)M}\mathrm{poly}(n, L)$.

operation	exponent per n
dictionary and canonical-layer preparation	μ
canonical witness output and complete cover state	$\mu(1 + D)$
all candidate-modulus energy evaluations	$\mu(\alpha + D/64 + d + \eta)$
selector and shared witness preprocessing	$\mu(1 + d + \eta)$
repeated raw prefix lookups	$\mu[1 + 2(d + \eta)]$
repeated outside lookups	$(1 - \mu)/2 + \mu(d + 2\eta)$
direct convolutions and common shifts	$\mu(d + \eta + 2D)$
high-deficiency dictionary matching	$1/2 - d_0\mu$
retained incidences and witness recovery	$1/2 - \mu g_*$.

(112)

A raw prefix list has at most 2^M times a fixed boundary factor, since $R \le 1$ and $Q_{\mathrm{ent}} < 1$. Enumerating it for every round and residue therefore gives the prefix-lookup line. Even scanning both outside lists once per residue costs at most $I \max_i D_i\, 2^{(n-M)/2-dM+O(1)}$, giving the outside line. Histogram convolution and shift scans cost $I \max_i Q_i^2 \mathrm{poly}(n, L)$. The energy row includes every candidate prime at all 64 stages; the prime sieve is smaller. Section I.4 certifies all these rates at $d < d_0$.

There are $M + 1$ layers and fixed numbers of boundary coordinates, count variables and prime factors. The counts have polynomial bit length in n, L. Comparisons and covers are exact; zero-difference tests are deterministic. Section I.2 supplies terminating input-independent compilation. Since $M = \mu n + O(1)$, lattice rounding adds $O(1)$ to logarithmic costs and gives uniformity for arbitrary binary input lengths.

I.4. Baseline analytic tools and numerical specification

This section records the baseline entropy tools, numerical menu, and cost calculation. The analytic inequalities are proved below. The contact values and subdivision counts in Section I.4.2 are historical reports for the older menu, whose full-domain replay is absent from the imported material. They remain the explicit premise of the background Theorem 1. The refined Theorem F2 uses the independently reconstructed certificate in Part II, Section 4. All terminating decimals denote exact rationals.

I.4.1. Entropy inequalities and rational logarithms

For $0 < q, p < 1$, the inequality $\ln y \le y - 1$ gives

$$q \ln \frac{p}{q} + (1-q) \ln \frac{1-p}{1-q} \le 0.$$

It follows that

$$H(x,y) \le -y \log_2 p - (x-y) \log_2 (1-p).$$

Endpoints follow by continuity. The function $H(x,y) = xh(y/x)$ is jointly concave: apply concavity of h with weights proportional to the two first coordinates, then multiply by their sum. Its restriction to an affine segment is therefore above the chord joining its endpoint values. These facts justify the affine bounds used below.

For a rational $1 \le v \le 2$, put $z = (v-1)/(v+1)$. Integrating the geometric series for $1/(1-z^2)$ gives

$$\ln v = 2 \sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2z^{65}}{65(1-z^2)}.$$

The tail bound follows by replacing its denominators by 65. Round each positive summand downward to a multiple of 2^{-120}; add $32 \cdot 2^{-120}$ and the displayed tail to obtain an upper bound. Apply the same calculation to $v = 2$. For any positive rational x, write $x = 2^k v$ with $1 \le v < 2$ and use $\log_2 x = k + \ln v / \ln 2$. The normalized numerator is nonnegative, so division by the opposite positive endpoint of the interval for $\ln 2$ is valid. For the profile calculation, round the resulting base-two logarithms outward to multiples of 10^{-14}. The scalar calculations in Section I.4.4 can retain the finer rational endpoints before this final rounding.

Entropy enclosures follow by rational interval evaluation of

$$H(x,y) = x \log_2 x - y \log_2 y - (x-y) \log_2 (x-y),$$

with $0 \log_2 0 = 0$ and $H(x,0) = H(x,x) = 0$. Also

$$h'(u) = \log_2 \frac{1-u}{u}, \qquad h''(u) = -\frac{1}{(\ln 2) u (1-u)}.$$

Integrating the second derivative from $1/2$, where $h' = 0$, yields $1 - h(1/2 + x) \ge 2x^2 / \ln 2$ on $|x| \le 1/2$. The same logarithm enclosures give $\ln 2 < 7/10$.

I.4.2. Recursive and trace-aggregated lookup contacts

Every decimal in the following table denotes an exact rational number. The first two rows use (42), the middle six use the trace-aggregated construction (43)–(53), and the last uses the direct occurrence circuit of Sections I.2.2–I.2.4. Within a row, both contacts concern one fixed circuit and one fixed routing rule independent of τ.

ρ	γ_1	Left τ	Upper bound	Right τ	Upper bound
1.3	.47	.24	.12004	.40	.16082
1.7	.47	.40	.15966	.45	.17177
2	.47	.45	.17078	.475	.1764
2.2	.47	.475	.17585	.49	.17901
2.4	.47	.49	.1786	.505	.18154
2.6	.47	.505	.18122	.52	.18393
2.8	.50	.52	.18347	.54	.18689

ρ	γ_1	Left τ	Upper bound	Right τ	Upper bound
3.1	.54	.54	.18616	.57	.19091
3.3	direct	.57	.19179	.76	.21588

The tails use

$$d_{\text{push}} \le .49973033201029, \qquad d_{\text{pull}} \le .88282502240658. \tag{113}$$

The claimed baseline contacts concern their full color domains. The following describes their reported certificate construction; it specifies acceptance rules but does not supply an independent replay of the historical run. For rational $0 < p < 1$, use the Gibbs supporting line

$$H(x, y) \le -y \log_2 p - (x - y) \log_2(1 - p). \tag{114}$$

Both negative logarithms are enclosed outward on the 10^{-14} grid by the rational logarithm procedure of Section I.4.1. Gibbs probabilities are rational multiples of 10^{-12} clamped to $[10^{-9}, 1 - 10^{-9}]$. Every such probability gives a valid supporting line. Near-stationary proposals reduce subdivision; acceptance uses the rational bounds.

An outward-rounded implementation normalizes to $x \in [1, 2]$, puts $q = (x - 1)/(x + 1)$, and uses

$$\ln x = 2 \sum_{j=0}^{31} \frac{q^{2j+1}}{2j + 1} + \mathcal{R}, \qquad 0 \le \mathcal{R} \le \frac{2q^{65}}{65(1 - q^2)}. \tag{115}$$

Keep each lower and upper operation outward on the 2^{-120} grid, including interval bounds on q and its powers. The same formula at $x = 2$ encloses $\ln 2$, and division gives rational bounds on $\log_2 x$. Expanding $H(x, y) = x \log_2 x - y \log_2 y - (x - y) \log_2(x - y)$ gives outward entropy bounds, with the zero terms defined continuously.

For the first-level contacts use boxes $[m_0, m_1] \times [z_0, z_1]$, starting from $[a, s] \times [a, \beta(s)]$ and tightening $z_0 \ge m_0$, $z_1 \le \beta(m_1)$. Choose Gibbs lines at a feasible center for the three positive inner entropies in $(1 - \tau)F_{\text{raw}} + \tau G$. Their weighted sum is affine in w. If its coefficient of w is nonnegative, replace w by either globally valid affine upper bound $\beta - m$ or C; if it is negative, replace it by one of $\beta - z, C - m, 0$. Choose the tightest at the center, but retain that same affine choice throughout the box. Each candidate remains a valid one-sided inequality throughout the feasible region.

One rational proposal for the inner point uses 48 bisections on (38). For $\tau = A/B$, the derivative sign is determined by the following rational comparison:

$$\left(\frac{(\beta - m - w)(C - w)}{w(m - C + w)}\right)^{B-A} \left(\frac{C - w}{w - \beta + z}\right)^{A}. \tag{116}$$

Clear denominators before comparing integer powers; use the unique endpoint when the interval is a singleton. This supplies the supporting-line parameters. In the branch $F = 0$, apply a Gibbs line directly to $G = H(C - \beta + z, w - \beta + z)$ and eliminate w with the same valid affine endpoints. Thus the two branches of $\max(0, F_{\text{raw}})$ are both covered.

The negative entropy $H(1 - m, \beta - m)$ has a lower linear chord in m. For $H(z, C - \beta + z)$, refine until $z_0 \ge (1 - \gamma)\beta(m_1)$, so every box vertex lies in its entropy domain. Its bilinear vertex interpolant is a lower bound by joint concavity. After these replacements every branch is multiaffine in (m, z); maxima and nonnegative sums preserve separate convexity. Its maximum on a rectangle therefore occurs at a vertex. Accept only if all four rational vertex bounds are

below the table contact; otherwise bisect the larger of $3(m_1 - m_0)$ and $z_1 - z_0$, with the first winning ties. This is the stated acceptance rule for the four first-level contacts. The fixed push branch in the two-level families uses the same method with $w_1 \le C_1/2$, projected lower bound $z \ge \max\{m, (1-\gamma_1/2)\beta\}$, and the additional globally valid upper endpoint $w_1 \le C_1/2$.

For the recursive branch set $v = w_1 - C_1/2 \ge 0$ and $C_2 = \gamma_1\beta/2$. Its full domain is

$$\max\{0, \beta - z - C_2, C_2 - m\} \le v \le \min\{C_2, \beta - m - C_2\}. \tag{117}$$

Put $c = (1-\gamma_1/2)\rho - 1 > 0$. Start from

$$m \in \left[\max\left\{a, \frac{(1-\gamma_1/2)a(\rho-1)}{c}\right\}, s\right], \quad z \in [a, \beta(s)], \quad v \in [0, \min\{C_2(s), \beta(s) - s - C_2(s)\}].$$

An empty range means that branch has no colors. On each box tighten

$$\begin{aligned} z_0 &\leftarrow \max\{z_0, m_0, \beta(m_0) - C_2(m_0) - v_1\}, \\ z_1 &\leftarrow \min\{z_1, \beta(m_1)\}, \\ v_0 &\leftarrow \max\{v_0, 0, \beta(m_0) - C_2(m_0) - z_1, C_2(m_1) - m_1\}, \\ v_1 &\leftarrow \min\{v_1, C_2(m_1), \beta(m_1) - m_1 - C_2(m_1)\}. \end{aligned} \tag{118}$$

All menu values have $\gamma_1\rho \le 2$, which justifies these monotone endpoint relaxations. Discard only empty boxes.

Choose Gibbs lines at a feasible center for the positive terms in $\tau F_{2,\mathrm{raw}} + (1-\tau)G_2$. The weighted sum is affine in w_2. Eliminate its coefficient using a valid affine upper endpoint D_1 or C_2 when nonnegative, and a lower endpoint $Q_2, C_2 - Q_1$, or 0 when negative. A rational stationary-point proposal on (45) compares

$$\left(\frac{(D_1 - w_2)(C_2 - w_2)}{w_2(Q_1 - C_2 + w_2)}\right)^A \left(\frac{C_2 - w_2}{w_2 - Q_2}\right)^{B-A} \gtrless 1 \tag{119}$$

for $\tau = A/B$. In the $F_2 = 0$ branch, put $y = w_2 - Q_2$; then

$$\max\{0, v - Q_1\} \le y \le \min\{v, w_1 - Q_1\}.$$

A Gibbs line for $H(v, y)$ and the appropriate signed affine endpoint eliminate the last color.

Use a lower chord for $H(1-m, \beta-m)$, lower bilinear interpolants for $H(z, D_1)$ and $H(C_2 + v, v)$, and positive Gibbs lines for the remaining entropies. Refine until $z_0 \ge (1-\gamma_1)\beta(m_1)$ to keep the first bilinear entropy valid at every vertex. The second is always valid since $v \ge 0$. For the trace correction put

$$\begin{aligned} D_{\min} &= (1-\gamma_1/2)\beta(m_0) - m_0 - v_1, \\ D_{\max} &= (1-\gamma_1/2)\beta(m_1) - m_1 - v_0, \\ (d-q)_{\min} &= m_0 - a - C_2(m_0) + v_0, \\ (m-q)_{\min} &= m_0 - C_2(m_0) + v_0, \\ (w-r)_{\min} &= C_2(m_0) + v_0 - s + m_0, \quad r_{\max} = s - m_0, \quad d_{\min} = m_0 - a. \end{aligned} \tag{120}$$

Verify $\beta(m_0) \ge s$, nonnegativity of the factors, and

$$(d-q)_{\min}(w-r)_{\min} \ge r_{\max} D_{\max},$$

$$r_{\max} m_1 D_{\max} \le d_{\min}(m-q)_{\min}(1-s-D_{\max}). \tag{121}$$

These imply (52) for every feasible color in the box. Only on such boxes replace F_1 by $\max\{0, F_{1,\mathrm{raw}} - \delta\}$ with (57); otherwise set $\delta = 0$. The subtraction stays inside the nonnegative maximum.

The resulting upper expression is separately convex in (m, z, v): each branch is multiaffine, and all outer sums and maxima have nonnegative weights. Evaluate every one of its eight vertices with rational arithmetic. Accept precisely when all are at most the table contact; otherwise bisect the coordinate with largest value among $3(m_1 - m_0), z_1 - z_0, 2(v_1 - v_0)$, with that tie order. Evaluate affine extensions on the relaxed-box vertices, with all negative entropy evaluations in their valid domains.

The baseline manuscript reported the following accepted nonempty leaf counts. These are historical report values; the refined checker does not reproduce or certify this older parameter menu.

ρ	Left contact: push + recursive	Right contact: push + recursive
2	$51 + 12447$	$51 + 10699$
2.2	$50 + 15549$	$50 + 14356$
2.4	$44 + 19271$	$45 + 18263$
2.6	$46 + 21263$	$45 + 21561$
2.8	$45 + 19175$	$46 + 19509$
3.1	$55 + 15764$	$55 + 16098$

The reported counts total 204538 accepted leaves in these twelve contacts, with positive trace corrections on 183117 recursive leaves. The reported four first-level counts are $176, 139, 143, 137$. For the two direct contacts start with $[a, s] \times [a, \beta(s)]$, tighten $z_0 \ge m_0, z_1 \le \beta(m_1)$, and use center-selected Gibbs lines for all positive terms in (24) and a lower chord for $H(1-m, \beta-m)$. If $m_0 > a$ and $\rho(m_0 - a)^2 \ge b(z_1 - a)$, subtract the constant $(7/5)(s - m_1)\max(0, z_0 - m_1)$ from the right degree; otherwise subtract zero. Evaluate both branches of L_ρ at all four vertices, accepting only below the contact. Bisect by the same $3(m_1 - m_0)$ versus $z_1 - z_0$ rule. The reported direct-contact counts are 175 and 183 accepted leaves. Their full-domain inequalities remain the explicit numerical premise of the background Theorem 1; these counts alone do not independently establish that premise. Finite cover factors and rounding are accounted for separately by ζ.

I.4.3. Whole-prefix compensation

Join each row's two contacts by its chord. On $[0, .24]$ use τd_{push} and on $[.76, 1]$ use $(1-\tau)d_{\mathrm{pull}}$. Denote this piecewise upper profile by f; at a boundary retain both one-sided values. Convexity of each fixed family's colored profile proves its chord throughout the interval.

Put

$$R = h(t), \quad Q = h(s) + H(s, a), \quad D_c = R - Q, \qquad .38309469039628 \le D_c \le .38309469039633,$$

$$K = \frac{875}{3042}, \qquad C = \frac{1}{2K} = \frac{1521}{875}, \qquad S_* = \frac{903}{2500} = .3612. \tag{122}$$

For $u = f(\tau)$, $v = f(1-\tau)$ and $\Delta = (2\tau - 1)D_c + u - v$, the finite rational comparison gives

$$u + v + K\Delta^2 \le S_* \qquad (0 \le \tau \le 1). \tag{123}$$

Here is the full reduction to the finite comparison. All chord values are in $[0, 1/4]$, so $|\Delta| \le .384 + .25 = .634$. The two partial derivatives with respect to u, v are positive because $2K \cdot .634 < 1$. Thus both upper chords may be substituted independently. The common refinement of their partitions has endpoints

$$0, .24, .40, .43, .45, .46, .475, .48, .49, .495, .505, .51, .52, .525, .54, .55, .57, .60, .76, 1. \tag{124}$$

On each of its nineteen intervals the expression is convex in τ; it is also convex in D_c. Check both interval endpoints at both rational endpoints for D_c. Thus there are thirty-eight one-sided profile comparisons, or seventy-six scalar rational inequalities. Their maximum is exactly

$$\frac{19776045461939145479265934608054 07}{54756000000000000000000000000000000} < .361167,$$

with margin below S_* equal to

$$\frac{18217380608545207340653919 4593}{54756000000000000000000000000000000} > .000033. \tag{125}$$

For the row-oriented core fraction q, let

$$\Phi(q, \tau) = q[\tau R + (1 - \tau)Q + u] + (1 - q)[(1 - \tau)R + \tau Q + v].$$

Writing $x = q - 1/2$ gives $\Phi = (R + Q + u + v)/2 + x\Delta$. Completing the square in (123) therefore proves

$$\Phi(q, \tau) \le v_*(q) := \frac{R + Q + S_*}{2} + C(q - 1/2)^2. \tag{126}$$

Indeed the difference is at most $-\Delta^2/(4C) + x\Delta - Cx^2 \le 0$. Since $\ln 2 < 7/10$, entropy curvature gives $1 - h(1/2 + y) \ge (20/7)y^2$. Apply this with $y = (1 - 2s)(q - 1/2)$ and use $C = (20/7)(1 - 2s)^2$ to obtain

$$v_*(q) \le \frac{R + Q + S_*}{2} + 1 - h\big(s + (1 - 2s)q\big). \tag{127}$$

The reflected orientation has the same entropy. Bounded core rounding changes the logarithmic cost only by a bounded quantity in the good-layer range used by the reduction. The compiler of Section I.2 consequently gives uniform endpoint degree caps with

$$\delta_U(q) = v_*(q) - [qR + (1 - q)Q], \qquad \delta_V(q) = v_*(q) - [qQ + (1 - q)R],$$

$$\deg_U \le O(1)2^{[\delta_U(q) + \zeta]M}, \qquad \deg_V \le O(1)2^{[\delta_V(q) + \zeta]M}, \qquad \zeta = 10^{-6}. \tag{128}$$

These give the per-endpoint enumeration bounds for the exact circuits.

I.4.4. Exact gain and complete reduction costs

Use the matching types of Section I.3, $S_* = 903/2500$, the full compilation allowance $\zeta = 10^{-6}$, and $d_0 = 1/50$. With the definitions there,

$$g_* = -\frac{1}{2} + \frac{1}{2}h(s) + \frac{1}{2}H(s,a) + H(1-s,t-a) - \frac{1}{2}h(t) - \frac{1}{2}S_* - \zeta.$$

The rational logarithm series in Section I.4.1 proves the safe bounds

$$\begin{aligned} g_* \ge g_L = \frac{2135335441677}{200000000000000}, \qquad D \le D_U = \frac{49809813343531}{10^{14}}, \\ \alpha \le \alpha_U = \frac{138991595816454}{10^{14}}, \qquad \eta \le \eta_U = \frac{559334743723}{10^{14}}. \end{aligned} \tag{129}$$

For example, outward intervals obtained from the same series are

$$\begin{aligned} &.49991595816452 < h(s) < .49991595816453, \\ &.00741117216645 < J < .00741117216646, \\ &.49809813343529 < D < .49809813343530, \\ &.00559334743721 < \eta < .00559334743722. \end{aligned} \tag{130}$$

Choose the rational balance

$$\mu = \frac{1}{2(1 + D_U + g_L)} = \frac{100000000000000}{301754962128739}.$$

Then

$$\mu(1 + D_U) = \frac{1}{2} - \mu g_L = E = \frac{149809813343531}{301754962128739} < 0.4964618056. \tag{131}$$

These rational upper bounds coincide. Their polynomially weighted sum has the same exponent. Also $3\mu < 1$ and $g_L < d_0$, so the dictionaries fit and the high-deficiency branch is strictly cheaper.

The following rates substitute the rational bounds (129) into (112). Decimal entries are rounded upward.

Operation	Upper rate per input coordinate
Retained incidences and witness recovery	E
Canonical output and complete residue-cover state	E
All candidate-modulus energy evaluations	.471671478
Selector and shared witness preprocessing	.339876217
Repeated prefix lookups	.348357717
Repeated outside lookups	.344637748
Direct cyclic convolution and common shifts	.338615680
High-deficiency dictionary matching	.493372106
Dictionary and canonical-layer preparation	.331394717

Each auxiliary rate is strictly smaller than E by exact rational comparison. The bound $h(1/5) < .74 < 1 - d_0$ places every nonempty good layer in $(1/5, 4/5)$ for sufficiently large M, as required by the core construction. The ledger includes local repetitions, modulus candidates, residue states, trie preprocessing, and witness recovery, with polynomial bit factors for all arithmetic.

I.5. Proof of Theorem 1

Sections I.2 and I.4 give exact typed circuits and common adjacency enumerators with total allowance $\zeta = 10^{-6}$. Their terminating input-independent compilation contributes C.

For an arbitrary integer instance, construct the dictionaries of Section I.3. If $N^{50} \le 2^{49M}$, use two-dictionary matching. Otherwise the canonical layers, typed selector, energy oracle, and harmonic covers serve every prefix, preserving actual representations at every outside residue. The balanced outside lists retain equal-sum witnesses.

Form original integer keys after undoing label flips, keeping round and residue-occurrence tags distinct. The signed circuit gives the nonnegative count of compatible equal-key pairs. A positive count yields a binary witness by bisection; if all counts vanish, completeness gives NO. This applies to signed weights, repetitions, zeros, and coincidences.

The ledger and (131) bound input-dependent work by $\mathrm{poly}(n, L)2^{En}$. Rounding and the boundary change only constant factors; the finite small-input range is handled exhaustively with a fixed polynomial coefficient. Adding C proves the theorem. □

Part II. Trace refinement and improved parameter setting

This part applies the exact circuit compiler and reduction proved in Part I to a stronger pooled-trace bound and a new rational parameter setting. Its numerical requirements are discharged by the reconstructed continuous-domain and scalar certificates described in Sections 4 and 9. Source revisions and attribution are recorded in the source map.

1. Theorem and dependencies

The background conditional composition in Theorem 1 has exponent

$$E_0 = \frac{149809813343531}{301754962128739} = .496461805587\ldots.$$

The change below gives

$$\boxed{E_2 = \frac{158153814247997}{318964058919134} = .4958358467845221\ldots < .495836.} \tag{XSS1}$$

The exponent decreases by approximately `.00062595880` per input coordinate. This result applies to arbitrary signed binary integers, with no promise on density, multiplicities, target, or the existence of a solution.

Theorem F2 (preserved two-level theorem). There is a uniform deterministic algorithm deciding Subset Sum on `n` indexed signed binary integers of total encoding length `L`, and returning an indexed witness on YES instances, using

$$C_{\mathrm{start},2} + \mathrm{poly}(n, L)2^{E_2 n} \tag{XSS2}$$

bit operations. The startup cost $C_{\mathrm{start},2}$ is finite and independent of the input. All polynomial degrees are fixed. This is a time theorem; it makes no simultaneous claim for the separate low-space algorithm.

The new argument changes the following concrete ingredients: the pooled-trace upper bound, the rational matching types used inside the reduction, and the finite menu of recursive lookup widths. The finite contact-cone lemma extends every contact beyond its original chord interval.

The argument then recomputes the entire profile and every outer cost line. The exact signed identities, fixed-seed compiler, canonical selection, energy evaluation, residue covers, exact integer keys, and witness lifting are proved in Part I. The argument below applies those constructions with the parameter changes justified in Section 6 and recomputes their cost in Section 7. It uses the new contact certificate directly and does not assume the older numerical menu in Section I.4.2.

A deterministic baseline cap

The finite circuit search need not finish before the instance can be answered. Let $t_{\mathrm{fast}}(x)$ be the bit-step count of the algorithm in Theorem F2 on an encoded instance x, including its startup search. Construct an ordinary meet-in-the-middle algorithm as follows. Split the indices into blocks of sizes $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$. Enumerate their exact subset sums, retaining the index mask with each entry, sort one indexed sum list, and search it for the target minus each sum in the other list. Each sum and comparison has polynomial bit cost in n, L. There are at most $2^{\lceil n/2 \rceil}$ entries per list, so this deterministic algorithm takes $t_{\mathrm{MITM}}(x) \le P_0(n, L)2^{n/2}$ bit operations for a fixed polynomial P_0. It returns the union of two disjoint index masks on YES instances and returns NO exactly when no complement occurs. Signed, zero and repeated weights require no separate assumption.

Run these two fixed exact programs by alternating their elementary bit steps, retaining both machine states. Their input copies and output buffers can be kept separate; copying the common input and the completed answer adds $O(L + n\log(n+1))$ bit operations. Return the first completed decision and, on YES instances, its indexed witness. If the faster program takes t steps, it finishes within at most $2t$ interleaved steps. Consequently the resulting deterministic algorithm satisfies

$$T(x) \le 2\min\{t_{\mathrm{fast}}(x), t_{\mathrm{MITM}}(x)\} + O(L + n\log(n+1)),$$

and hence, for fixed polynomials P_0, P_1,

$$T(n, L) \le 2\min\left\{C_{\mathrm{start},2} + P_1(n, L)2^{E_2 n}, P_0(n, L)2^{n/2}\right\} + O(L + n\log(n+1)).$$

This preserves the proved faster exponent while bounding every instance by the meet-in-the-middle schedule, even if its circuit search has not yet completed. The corollary is about time and exact indexed output; it adds no simultaneous space guarantee and no practical implementation claim for the finite circuit compiler.

The same interleaving applies to Theorem F3 after Part III. In that application, replace $C_{\mathrm{start},2}, E_2$ in the displayed bound by $C_{\mathrm{start},3}, E_3$; the indexed-output proof and the common input/output overhead are unchanged.

2. New types and the exact circuit families

Set

$$s = \frac{7}{50} = .14, \quad t = \frac{9}{20} = .45, \quad a = \frac{13}{200} = .065, \quad b = \frac{3}{40} = .075, \qquad a + b = s. \tag{XSS3}$$

Write `H(c,x)=c log_2 c-x log_2 x-(c-x)log_2(c-x)`, with its continuous endpoint values. The outer color domain is

$$a \le m \le s, \qquad \beta = a + \rho(m-a), \qquad m \le z \le \beta,$$

$$L_0 = \max\{0, H(t,z) + H(1-t, \beta - z) - H(1-m, \beta - m)\}. \tag{XSS4}$$

Use the following fixed recursive parameter triples. The third entry is the second lookup fraction; all original half-lookup families are retained.

$$\begin{aligned}(\rho, \gamma_1, \gamma_2) \in \{&(1.3, .47, .50), (1.7, .47, .50), (2.2, .50, .50), (2.6, .54, .50),\\ &(2.8, .58, .50), (3, .60, .50), (3.15, .664, .56), (3.29, .664, .56),\\ &(3.4, .62, .50), (3.43, .664, .56), (4, .65, .50)\}.\end{aligned} \tag{XSS5}$$

All decimals in the mathematical specification are exact rationals. For every family, `beta<=.365<t<1-t`. Its first inner lookup is `c=gamma_1 beta`. The required containment condition `beta-z<=c` holds on the full color domain. Its defect `beta-m-c` is affine in `m`; at `m=a` it equals `-gamma_1 a<0`. At the other endpoint `m=s`, the pairs `(beta-s,c)` for the eight original families are respectively

$$(.0225, .076375),\ (.0525, .090475),\ (.09, .115),\ (.12, .1404),$$

$$(.135, .1595),\ (.15, .174),\ (.18, .1984),\ (.225, .23725).$$

The three added families have pairs

$$(.16125, .20003), \qquad (.17175, .207002), \qquad (.18225, .213974).$$

Their positive containment slacks are `.03878`, `.035252` and `.031724`. The scalar checker verifies all eleven families using exact rationals.

The first inner color and occurrence exponent are

$$Q_1 = \beta - z, \quad D_1 = c - Q_1, \qquad \max\{Q_1, c-m\} \le w_1 \le \min\{\beta - m, c\},$$

$$F_{1,\mathrm{raw}} = H(\beta - m, w_1) + H(m, c - w_1) - H(z, D_1). \tag{XSS6}$$

Use the exact second-level routing

$$Q_2 = c - w_1, \qquad C_2 = \max\{\gamma_2 c, Q_2\}, \qquad D_2 = C_2 - Q_2. \tag{XSS7}$$

On the push branch `w_1<=(1-gamma_2)c`, use the old valid profile upper bound

$$P_{\mathrm{push}} = \tau L_0 + (1-\tau)H(b, m-a) + (1-\tau)\max\{0, F_{1,\mathrm{raw}}\} + \tau H(D_1, w_1 - Q_1). \tag{XSS8}$$

On the recursive branch, write `v=w_1-(1-gamma_2)c>=0`, so `C_2=gamma_2 c` and `D_2=v`. The complete domain is

$$\max\{0, \beta - z - (1-\gamma_2)c, \gamma_2 c - m\} \le v \le \min\{\gamma_2 c, \beta - m - (1-\gamma_2)c\},$$

$$\max\{Q_2, C_2 - Q_1\} \le w_2 \le \min\{D_1, C_2\}. \tag{XSS9}$$

Put

$$F_2 = \max\{0, H(D_1, w_2) + H(Q_1, C_2 - w_2) - H(w_1, D_2)\},$$

$$G_2 = H(D_2, w_2 - Q_2). \tag{XSS10}$$

Why the generalized second width is an exact circuit

The positive-containment identity in Section I.2.5 permits any lookup containing its query. Here `0<=Q_2<=C_2<=c`. At finite integer sizes, choose `max(ceil(gamma_2 |C|),|Q_2|)`; it remains at most `|C|`. The same complement/transposition therefore gives (XSS10) for both `gamma_2=1/2` and `gamma_2=14/25`.

Put `b_2=(1-gamma_2)c`. Eliminating the push color gives the interval

$$\max\{Q_1, c - m, 0\} \le w_1 \le \min\{\beta - m, b_2\},$$

and the necessary projection inequalities `m>=gamma_2 c` and `z>=beta-b_2`. The acceptance bound uses `b_2` as its upper endpoint; it does not replace that endpoint by `gamma_2 c`. The recursive domain in (XSS9) is exactly the complementary closed branch. At `v=0`, `Q_2=C_2`, `G_2=0`, and

$$F_2 = H(D_1, Q_2) = H(D_1, w_1 - Q_1).$$

Thus the unmerged degree contributions agree at the routing boundary. The complete upper profiles need not agree there: the recursive profile uses the sharper trace bound below, whereas (XSS8) retains the older valid upper bound. Both closed domains are certified. The first-lookup trace and its aggregated integer coefficient are unchanged by the new second width.

The new profile replaces the old first-level right contribution by the trace bound proved next. The finite signed matrix is still the same exact aggregated circuit from baseline (47); no positive cover substitutes for a signed identity.

Also retain the ordinary direct signed lookup family `rho=4`. Its profile is bounded by

$$P_{\mathrm{dir}} = \tau L_0 + (1 - \tau)\{H(b, m - a) + H(\beta - m, z - m) - \delta_{\mathrm{dir}}\}. \tag{XSS11}$$

On a rectangle where `m_0>a` and `rho(m_0-a)^2>=b(z_1-a)`, use `delta_dir=(7/5)(s-m_1) max(0,z_0-m_1)`; otherwise use zero. This is the existing direct distinct-gate bound, applied at the new types.

3. The new pooled-trace lemma

Fix one outer and first-inner occurrence and write

$$d = m - a, \quad r = s - m, \quad q = c - w_1, \quad D = \beta - m - w_1, \qquad B = H(b, d).$$

The first-inner domain gives `0<=q<=m` and `D>=0`. When `beta>=s`, define

$$A(g) = H(b, g) + H(s - g, q) + H(\beta - s, w_1 - g) - H(z, D_1),$$

$$J(u) = H(b, d+u)+H(m+u, q)+H(1-s, D-u)+H(1-m-D, w_1)-H(1-m, w_1+D)-H(z, D_1). \tag{XSS12}$$

Both functions are concave in their scalar argument. Their intervals are

$$I_g = [\max\{0, w_1 - (\beta - s)\}, \min\{r, w_1\}], \qquad I_u = [0, \min\{r, D\}]. \tag{XSS13}$$

All entropy arguments are feasible. In particular, for this menu `D<=beta<=.365<1-s=.86`. Put `A_*=max_{I_g}A` and `J_*=max_{I_u}J`. Then the same exact aggregated circuit has a right-degree upper exponent

$$R^{\mathrm{trace}} \le G_2 + B_{\mathrm{tr}},$$

$$\boxed{B_{\mathrm{tr}} = \min\left\{B + \max\{0, F_{1,\mathrm{raw}}\}, \max\{B, A_*, J_*\}\right\}.} \tag{XSS14}$$

When `beta<s`, define `B_tr=B+max(0,F_{1,raw})`. Thus on every recursive color the complete profile is bounded by

$$P_{\mathrm{rec}} = \tau L_0 + (1-\tau)B_{\mathrm{tr}} + \tau F_2 + (1-\tau)G_2. \tag{XSS15}$$

Proof of the trace lemma

Use the exact pooled-histogram expression (51) in the baseline. Its histogram coordinates are `j,e,f`. Set `k=q-e` and `g=f-k`. The three branches of the pooled count are `0,C_1,C_0+C_1`.

For the zero branch, every feasible trace `K` has at least one completion `U=Y_1 union V`, where `V` has `dN` elements of `Y_0`. Select one completion for each trace. Distinct traces have different `U cap C`, and therefore different completions. Hence there are at most `binom(bN,dN)` traces, giving exponent `B`. Equivalently, the feasibility constraints give `0<=k<=d` and `0<=d-k<=j-f`; a fixed outside completion injects the `binom(fN,kN)` trace choices into the `binom(bN,dN)` original choices. This argument does not require `q<=d`.

For the C_1 branch, fix the outer universe B of size βN and its prescribed-one set Y_1 of size aN. Since $\beta \ge s$, choose nested prescribed-zero sets $Z_j \subseteq Z_b \subseteq B \setminus Y_1$ of sizes jN and bN. A counted pair (C, K) has $C \cap Y_1 \subseteq K \subseteq C \cap (Y_1 \cup Z_j)$, $|K| = qN$, and a completion $U = Y_1 \cup V$ with $V \subseteq Z_j$, $|V| = dN$, and $U \cap C = K$. Enlarging Z_j to Z_b leaves C, K, U unchanged, so this is an injection into the union of the feasible histograms at $j = b$. The restriction that B owns U was already relaxed in the histogram upper bound; the injection need not preserve an ownership assignment. The target union has only polynomially many values of the new zero intersection count. Its largest entropy exponent therefore bounds the original one, while the denominator $H(z, D_1)$ remains unchanged. With $k = q - e$ and $g = f - k$, the exponent at $j = b$ is

$$H(a, q-k) + H(b, k) + H(b-k, g) + H(\beta - s, w_1 - g) - H(z, D_1).$$

The entropy chain identity and Vandermonde give

$$H(b, k) + H(b-k, g) = H(b, g) + H(b-g, k),$$

$$H(a, q-k) + H(b-g, k) \le H(s-g, q).$$

Completion implies `0<=g<=r`, and the remaining entropy domains give `I_g`. Thus this branch is at most `A_*`. The change from the earlier argument is to sum over `k` before maximizing over `g`.

For the `C_0+C_1` branch set

$$u = j - f - (d - k), \qquad j = d + u + g.$$

Feasibility gives `0<=u<=min(r,D)`. The numerator exponent rewrites as

$$H(b, d+u) + H(d+u, k) + H(a, q-k) + H(r-u, g) + H(1-s, D-u) + H(1-s-D+u, w_1-g).$$

The two negative terms are `H(1-m,w_1+D)` and `H(z,D_1)`. Vandermonde in `k` gives `H(m+u,q)`; independently, Vandermonde in `g` gives `H(1-m-D,w_1)`. The resulting expression is `J(u)`. Maximize over `I_u`, and combine the three branches. Polynomially many histogram choices do not change normalized exponents. Taking the minimum with the old bound for the same circuit proves (XSS14).

A useful closed corollary

If, in addition, `0<=q<=d`, `w_1>=r`, and

$$d(m-q)(w_1-r) \geq rmD, \tag{XSS16}$$

then the same closed correction as before is valid:

$$B_{\mathrm{tr}} \leq B + \max\{0, F_{1,\mathrm{raw}} - \Delta\}, \qquad \Delta = H(1-m, D) - H(1-s, D). \tag{XSS17}$$

Indeed, for `rD>0`,

$$A'(r) = \log_2 \frac{d(m-q)(w_1-r)}{rmD} \geq 0,$$

so concavity gives `A_*=A(r)`. Also

$$J'(0) = \log_2 \frac{rmD}{d(m-q)(1-s-D)} \leq 0,$$

because `(1-s-D)-(w_1-r)=1-beta>=0`. Thus `J_*=J(0)=B+F_{1,raw}-Delta`. The entropy-integral comparison in the baseline gives `J(0)>=A(r)`. If `r=0` or `D=0`, the interval is a singleton or `Delta=0`, and the old bound handles the boundary directly.

This guard is weaker than the old first guard: `(b-q)w_1>=r(beta-a-q)` is equivalent to `(d-q)(w_1-r)>=rD`; multiply by `m` and use `d(m-q)-m(d-q)=aq>=0` to obtain (XSS16). The old second guard follows automatically. The continuous bound (XSS14) remains available even when the closed guard fails.

4. Full-domain contacts

The following bounds concern one fixed exact family in each row. Every push branch, recursive branch, nonnegative maximum, and legal color is included. A pair `(tau,U)` means that the full normalized colored profile at `tau` is at most the exact rational `U`.

Exact family (rho,gamma_1,gamma_2)	Certified contacts (tau,U)
Recursive (1.3,.47,.5)	(.24,.14310), (.40,.19000)
Recursive (1.7,.47,.5)	(.40,.18545), (.44,.19560)
Recursive (2.2,.50,.5)	(.43,.19229), (.46,.19877)
Recursive (2.6,.54,.5)	(.45,.19669), (.48,.20234)
Recursive (2.8,.58,.5)	(.47,.20032), (.49,.20396)
Recursive (3,.60,.5)	(.48,.20214), (.49,.20384), (.50,.20556), (.51,.20730)
Recursive (3.15,.664,.56)	(.48,.20200), (.49,.20371), (.50,.20544)
Recursive (3.29,.664,.56)	(.49,.20377), (.50,.20538), (.51,.20700)
Recursive (3.4,.62,.5)	(.50,.20561), (.51,.20706), (.54,.21148), (.58,.21758)
Recursive (3.43,.664,.56)	(.50,.20545), (.51,.20695), (.52,.20846)
Recursive (4,.65,.5)	(.55,.21218), (.60,.21801), (.65,.22429)
Direct rho=4	(.60,.21873), (.76,.22760)

For one fixed family, the supremum of tau L+(1-tau)R over its colors is convex in tau. The trace bound is independent of tau. Hence a chord between consecutive contacts bounds the same family's whole intervening interval. No chord joins unrelated circuit families.

Rational proof method for the contacts

Use the baseline Gibbs inequality, for any rational 0<p<1,

$$H(x,y) \le -y\log_2 p - (x-y)\log_2(1-p). \tag{XSS18}$$

Positive entropies are replaced by these affine majorants. Negative entropies use lower linear or multilinear vertex interpolants, valid by concavity. If a relaxed box has an invalid corner for H(z,D_1), use the safe lower bound zero for that entire negative term on that box.

For (XSS14), apply the positive-entropy Gibbs bounds separately to A(g) and J(u). Each resulting expression is affine in its extra scalar. According to its coefficient's sign, substitute any one globally valid upper or lower interval endpoint from (XSS13). This removes g and u without adding a subdivision dimension. The negative terms remain the same H(z,D_1) and H(1-m,beta-m) already handled by interpolation.

Similarly eliminate w_2 using (XSS9), treating both branches of F_2. The resulting functions are separately convex in (m,z,v), so their maxima are attained at the eight box vertices. The old trace guard or the stronger guard (XSS16) may also be used on an entire box, with safe constant subtraction

$$\delta = \frac{7}{5}\frac{(s-m_1)D_{\min}}{1-m_0}$$

inside the nonnegative maximum. Maximize each alternative bound over the entire box's vertex set before taking the smaller resulting number. A pointwise minimum of separately convex bounds need not be separately convex, so taking a minimum independently at the vertices would not justify acceptance. The push and direct branches use the analogous two-coordinate calculation.

Start from the whole domains in (XSS4), (XSS6), and (XSS9), tighten only by necessary affine inequalities, and discard only empty boxes. For recursive boxes, bisect the coordinate maximizing 3(m_1-m_0), z_1-z_0, 2(v_1-v_0), in that tie order. For two-coordinate boxes

omit the third entry. The family `(1.3,.47)` has an empty recursive branch; its complete domain is handled by the push calculation.

Several families have `gamma_1 rho>2`. No sign of the generalized affine function `gamma_2 c-m` is copied from the old half-lookup certificate. Every affine interval extremum is evaluated at both relevant endpoints. For example, the lower bound for `m-q=m-gamma_2 c+v` is

$$\min\{m_0 - \gamma_2 c(m_0), m_1 - \gamma_2 c(m_1)\} + v_0. \tag{XSS19}$$

The old auxiliary shortcut `gamma_1 rho<=2` is unused. Neither it nor a sign restriction on `gamma_1 gamma_2 rho` is required by the exact circuit construction or this checker.

Rational Gibbs probabilities can be proposed by approximate stationary points, rounded to multiples of `10^{-12}` and clamped to `[10^{-9},1-10^{-9}]`. Acceptance uses only exact rational comparisons with outward logarithm enclosures. Normalize a positive rational into `[1,2]`, put `z=(x-1)/(x+1)`, and use

$$\ln x = 2\sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \qquad 0 \le \mathcal{R} < 2^{-100}. \tag{XSS20}$$

The tail follows from `z<=1/3`. Products and divisions are rounded outward on the `2^{-128}` grid. Divide by an independently enclosed `ln 2` and round base-two logarithms outward to multiples of `10^{-14}`. Approximate optimization chooses useful rational planes; it does not accept boxes.

The contact subdivisions are finite computations performed before the input algorithm. Their scalar variables do not add an input-dependent enumeration or optimization step.

The executable continuous-domain checker completed all 60 push/recursive calculations and both direct calculations with exact rational acceptance. Its full-run receipt records 248,189 accepted leaves, 496,320 subdivision nodes, two empty recursive domains, and maximum depth 29. The accepted leaf counts are given below in the same contact order as the preceding table. Each pair is `(push leaves, recursive leaves)`; zero means an empty recursive domain.

Family `(rho,gamma_1,gamma_2)`	Accepted leaf counts by contact
`(1.3,.47,.5)`	`(177,0)`, `(25,0)`
`(1.7,.47,.5)`	`(123,119)`, `(120,210)`
`(2.2,.50,.5)`	`(80,17319)`, `(80,14139)`
`(2.6,.54,.5)`	`(56,15591)`, `(59,13576)`
`(2.8,.58,.5)`	`(52,11670)`, `(52,10663)`
`(3,.60,.5)`	`(48,10249)`, `(48,9932)`, `(49,9562)`, `(49,9402)`
`(3.15,.664,.56)`	`(47,6760)`, `(47,6513)`, `(47,6382)`
`(3.29,.664,.56)`	`(46,6484)`, `(46,6405)`, `(47,6397)`
`(3.4,.62,.5)`	`(48,9331)`, `(49,9163)`, `(49,9386)`, `(49,9580)`
`(3.43,.664,.56)`	`(45,6782)`, `(46,6811)`, `(46,7011)`
`(4,.65,.5)`	`(47,8278)`, `(53,8872)`, `(60,9427)`
Direct `rho=4`	`161`, `224`

The smallest exact acceptance margin over all nonempty leaves is

$$\frac{136285389119}{128000000000000000000} > 10^{-9}.$$

The receipt binds the complete contact menu to the checker's exact source bytes. It is a record of the completed calculation; rerunning `--all` reconstructs the entire continuous-domain

proof. Approximate optimization only proposes rational planes and never accepts or discards a box. The verification note proves the coverage and acceptance rules.

5. The whole reflected profile

Finite contact cones

For a finite exact colored circuit, Section I.2.1 deletes gates with a zero factor and defines positive integer color degrees $A_j, B_j \geq 1$. Its colored volume is $\Upsilon(\tau) = \sum_j A_j^\tau B_j^{1-\tau}$. For a contact weight $\alpha \in (0,1)$, put

$$\kappa = \max\left\{\frac{\tau}{\alpha}, \frac{1-\tau}{1-\alpha}\right\} \geq 1.$$

Both exponents on the left below are at most the corresponding exponents on the right, and both degree bases are at least one. Therefore

$$A_j^\tau B_j^{1-\tau} \leq \left(A_j^\alpha B_j^{1-\alpha}\right)^\kappa, \qquad \Upsilon(\tau) \leq \sum_j \left(A_j^\alpha B_j^{1-\alpha}\right)^\kappa \leq \Upsilon(\alpha)^\kappa.$$

The last inequality follows from $\sum_j x_j^\kappa \leq (\sum_j x_j)^\kappa$ for nonnegative x_j and $\kappa \geq 1$. This is an exact finite-circuit inequality, including the entire color sum.

Consequently, a contact (α, U) supplies the two linear upper bounds

$$\frac{U(1-\tau)}{1-\alpha} \quad (0 \leq \tau \leq \alpha), \qquad \frac{U\tau}{\alpha} \quad (\alpha \leq \tau \leq 1).$$

If the finite seed contact has additive exponent error ϵ, these bounds have error at most $\kappa\epsilon$. All current contact weights lie in $[6/25, 19/25]$, so $\kappa \leq 25/6$ uniformly. Section 6 charges this constant amplification to the template allowance. No additional continuous contact is assumed, and the contact checker and its menu are unchanged.

Add the two exact endpoint circuits, valid for every τ:

$$f_{\mathrm{push}}(\tau) = .58414212731239\,\tau, \qquad f_{\mathrm{pull}}(\tau) = .85319345691908\,(1-\tau). \tag{XSS21}$$

The constants enclose $H(t,a) + H(1-t,b)$ and $H(1-s,t-a)$, respectively. Let f be the minimum of these lines, all eligible consecutive same-family contact chords from Section 4, and the contact-cone pieces above. A chord is eligible only inside its certified interval.

This extended envelope is continuous. The global minimum of the finitely many contact cones is continuous. At either endpoint of any added chord, the cone of that endpoint's contact has the same value, so adding the restricted chord cannot introduce a jump. The scalar checker verifies every resulting junction equality exactly. It still checks both one-sided evaluations for complete endpoint coverage.

Set

$$D_c = h(t) - h(s) - H(s,a), \qquad .26905132960667 \leq D_c \leq .26905132960673, \quad K = \frac{875}{2592},$$

$$U = f(\tau), \quad V = f(1-\tau), \qquad \Theta = (2\tau-1)D_c + U - V.$$

Then the complete scalar inequality is

$$\boxed{U+V+K\Theta^2\le S,\qquad S=.41077126.} \tag{XSS22}$$

All selected values are below $1/4$, so $|\Theta|<.52$ and the left side is increasing in each of U,V. It is convex in τ on every common affine refinement interval and is convex in D_c. The breakpoints of f are exactly

$$\begin{aligned}\mathcal{B}=\{0,\ &\frac{2425000000000}{9700570910413},\frac{5672}{14453},\frac{2}{5},\frac{288877}{673855},\\ &\frac{43}{100},\frac{3759}{8300},\frac{2114877}{4502050},\frac{47}{100},\frac{177893}{370800},\\ &\frac{12}{25},\frac{49}{100},\frac{99}{200},\frac{1}{2},\frac{607}{1200},\frac{51}{100},\frac{13}{25},\\ &\frac{257244}{494425},\frac{27}{50},\frac{308143}{561610},\frac{11}{20},\frac{3}{5},\frac{5709}{9355},\\ &\frac{16693148922977}{22715773922977},1\}.\end{aligned} \tag{XSS23}$$

Adjoin their reflections and check every interval endpoint from both sides and both endpoints of the interval for D_c. The executable scalar certificate constructs this full common refinement and checks 160 exact rational comparisons. Their maximum is

$$\frac{10515744200139750970543135213463}{256000000000000000000000000000000}=.41077125781795903\ldots<.41077126. \tag{XSS24}$$

The maximum occurs at $\tau=99/200$ and its reflection $101/200$. The exact positive margin is

$$\frac{55860249029456864786537}{256000000000000000000000000000000}>2\cdot10^{-9}.$$

The numerical values of S for different matching types are not directly comparable. The running-time comparison uses the rebalanced reduction below, with all type-dependent entropy terms recomputed.

6. Unequal cores, rational types, and the finite compiler

Put `R=h(t)`, `Q=h(s)+H(s,a)`. For a fraction `lambda` of row-oriented core coordinates, the endpoint objective is

$$\Phi(\lambda,\tau)=\lambda[\tau R+(1-\tau)Q+U]+(1-\lambda)[(1-\tau)R+\tau Q+V].$$

Completing the square using (XSS22) gives

$$\Phi(\lambda,\tau)\le\frac{R+Q+S}{2}+\frac{1296}{875}(\lambda-1/2)^2. \tag{XSS25}$$

The entropy curvature bound is valid for these types:

$$\frac{1296}{875}(\lambda-1/2)^2=\frac{20}{7}(1-2s)^2(\lambda-1/2)^2\le1-h(s+(1-2s)\lambda). \tag{XSS26}$$

This follows from `1-h(1/2+x)>=2x^2/ln 2>20x^2/7` on the whole binary entropy domain. Thus the complete mixed-core profile hypothesis holds.

The new rational types require a divisibility change. Use `H_0=1000`, take a sufficiently large fixed `k` divisible by `1000`, and use microblocks of length

$$N_0 = k^2 + 1000k.$$

The grouping discrepancies are still at most `12k`. The added matching dummy positions have target row-one, query-one, and query-zero counts `450k,65k,75k`, respectively. These have ample slack for the discrepancies and for fitting query ones and zeros into the corresponding row symbols. Both the description overhead `2 log_2(k+1)/k` and the new padding factor `1+1000/k` have the same limiting behavior required by the compiler.

Use core-fraction grid size 10000. Its quadratic interpolation loss is at most $(1296/875)/(4 \cdot 10000^2) < 3.71 \cdot 10^{-9}$. Set the complete compilation allowance to $\zeta = 10^{-8}$: allocate less than $2.5 \cdot 10^{-9}$ to the stored-template error, less than $2.5 \cdot 10^{-9}$ to count descriptions and dummy-padding inflation, and less than $5 \cdot 10^{-9}$ to interpolation including its padding factor.

The stored-template allowance includes both lookup ceilings, the seed's color and histogram factors, contact-cone amplification, and finite-depth rebalancing. Choose the pre-cone seed error below $3 \cdot 10^{-10}$. The exact bound $\kappa \le 25/6$ makes its post-cone contribution less than $1.25 \cdot 10^{-9}$. Then choose the common amplification depth large enough that its further loss is below $1.25 \cdot 10^{-9}$. Their sum is strictly below the allotted $2.5 \cdot 10^{-9}$.

For the lookup-rounding step use the full pooled histogram expression (51) and the unmerged degree identities, whose entropy terms are continuous on compact feasible domains. Limits of rounded feasible colors remain in the retained closed routing domains. The same uniform $o(1)$ argument as Section I.2.8 applies to the added rational widths. The finite cone inequality then transfers the already rounded seed bound to every weight; it does not require continuity of a simplified upper profile at a routing boundary.

Choose the fixed microblock length first, then the common amplification depth, large enough for all these bounds. The interpolation has a strict reserve, so Sections I.2.13–I.2.16 provide such fixed choices. Enlarging these data increases the finite startup constant; no practical startup bound is asserted. Every exact cover, aggregation identity, grouping restriction and stored adjacency list remains valid. Dovetailing over these finite data and checking exact integer matrix equality gives a terminating uniform startup search.

At input length, use the same count descriptions, tensor products of stored adjacency lists, actual gate names, and output-sensitive incidence enumeration. The additional scalar entropy optimizations in the proof are absent from this input-time implementation.

7. Retyping the reduction and the full outer ledger

The three-dictionary reduction is unchanged except for its rational types and block scale. Its low-deficiency canonical layers satisfy `1/5<k/M<4/5` exactly as before. Since the new `s=.14<1/5`, both integral cores remain positive in the required range. Replacing `H_0=500` by `1000` gives integral counts for all new types and only a fixed boundary overhead.

The canonical transformed pattern density and type-dependent constants are

$$u = t - a + b = .46, \qquad D_{\mathrm{can}} = H(1-u,a) + H(u,b),$$

$$\eta = 1 - h(u), \quad \alpha = 1 + h(s) - s, \quad J = 1 - H(s,a) - H(1-s,t-a). \tag{XSS27}$$

The canonical support bound, expectation identity, and `K/T` ratio are the same binomial/entropy identities at these new rational arguments. The initial difference table in the

energy oracle also needs its own retyping check. It has at most $\sum_{j\le 2sM+O(1)} \binom{M}{j}2^j$ entries. Since $2s < 2/3$, its exponent is at most $h(2s) + 2s$; exact entropy enclosures give

$$h(2s) + 2s < 1.135451 < 1.444238 < 1 + h(s) - s = \alpha.$$

Thus it fits the same energy-transform budget. The stronger baseline comparison $h(2s)+2s < 1$ is not required and does not hold at these refined types. The channel-three contraction bound still applies because $s < 1/3$. The scalar certificate checks these two domains, every outer lookup's containment condition, the dummy-padding inequalities, and the new compiler grid allowance.

In particular,

$$h(u) + D_{\mathrm{can}} = 1 + h(s) - J.$$

The gain is recomputed from its full formula:

$$g_* = -\frac{1}{2} + \frac{1}{2}h(s) + \frac{1}{2}H(s,a) + H(1-s,t-a) - \frac{1}{2}h(t) - \frac{1}{2}S - \zeta. \tag{XSS28}$$

Outward rational entropy evaluation as in (XSS20) proves

$$\begin{aligned} D_{\mathrm{can}} \le D_U &= \frac{58153814247997}{10^{14}}, \\ \alpha \le \alpha_U &= \frac{144423881164287}{10^{14}}, \\ \eta \le \eta_U &= \frac{231078058989}{5\cdot 10^{13}}, \\ g_* \ge g_L &= \frac{132821521157}{10^{13}}. \end{aligned} \tag{XSS29}$$

Keep the dictionary-deficiency threshold `d_0=1/50`. Choose

$$\mu = \frac{1}{2(1 + D_U + g_L)} = \frac{50000000000000}{159482029459567},$$

$$M = 1000 \left\lfloor \frac{\mu n}{1000} \right\rfloor. \tag{XSS30}$$

Then `3mu<1`, so the three dictionaries fit, and `g_L<d_0`. The two binding input-dependent rates are exactly

$$\mu(1 + D_U) = \frac{1}{2} - \mu g_L = \frac{158153814247997}{318964058919134} = E_2. \tag{XSS31}$$

Here is the complete ledger, with all displayed decimal caps rounded upward:

Operation	Rate per original input coordinate
Retained signed-circuit incidences and witness recovery	`E_2`
Canonical outputs and complete residue-cover state	`E_2`
All candidate-modulus energy evaluations	`<.463358446`
Selectors and shared witness preprocessing	`<.321234175`
Repeated prefix lookups	`<.328953402`
Repeated outside lookups	`<.352410683`
Direct cyclic convolution and common shifts	`<.372361028`

Operation	Rate per original input coordinate
High-deficiency dictionary matching	`<.493729702`
Dictionary and canonical-layer preparation	`<.313514947`

These are exact substitutions into, respectively, the full existing input-cost expressions

$$\begin{gathered}\mu(1+D_U),\quad \mu(d_0+\eta_U+\alpha_U+D_U/64),\quad \mu(1+d_0+\eta_U),\\ \mu[1+2(d_0+\eta_U)],\quad (1-\mu)/2+\mu(d_0+2\eta_U),\quad \mu(d_0+\eta_U+2D_U),\\ 1/2-d_0\mu,\quad 1/2-\mu g_L,\quad \mu.\end{gathered}\tag{XSS32}$$

The modulus candidates, full residue state, repetitions of the local covers, outside scans, canonical witnesses, integer-key bit lengths, and large-deficiency fallback are included. The change in `H_0` contributes only fixed padding and rounding costs. Handle the fixed finite small-input range exhaustively, with polynomial dependence on `L`. This yields (XSS2) under the same uniformity and exact-witness contract as the baseline.

8. Scope

The main new lemma replaces a constrained three-histogram maximization by two concave scalar upper bounds and weakens the old closed-correction guard. The second lookup also admits the exact rational fraction `14/25`, with complete color domains and the same trace aggregation. The finite contact-cone inequality extends the certified weights to a continuous whole-interval envelope, with its seed amplification explicitly charged. Retyping and the enlarged circuit menu convert these local changes into the stated general deterministic time exponent. The claim is not restricted to a favorable class of instances or to a single active color.

The startup search remains a large input-independent constant, with no practical numerical bound supplied here. The result does not combine this time exponent with the repository's separate small-space exponent, and it does not change the unrelated matching exponent `4/5`.

9. Reproducible numerical certificates

`verify_subset_sum_contacts.py` reconstructs the complete contact proof of Section 4. Run it with `--all` to repeat all 62 branch calculations; `--jobs` only controls parallelism. Every domain tightening and accepted vertex bound uses exact rational arithmetic. The receipt records the complete menu, the accepted bounds and margins, and the source-code and menu fingerprints. Receipt binding can be checked separately, without confusing that inexpensive provenance check with a full recomputation.

`verify_subset_sum_scalars.py`, run with `--base-two-level`, independently reconstructs this part's preserved continuous contact/chord/cone envelope and proves its entropy enclosures with rational logarithm series, checks all 160 reflected-profile comparisons, and checks every operation in the outer ledger. It reproduces the exact maximum (XSS24) and exponent (XSS31) without floating-point acceptance. These two programs discharge the numerical requirements of Theorem F2. The circuit compiler, reduction, and witness recovery are proved by the mathematical arguments in this manuscript. The verification note gives the complete commands and the separate scope of the finite energy-tensor check.

Part III. A third positive-containment lookup

The final-source contact receipt covers all 16 new branches in a completed resumable replay under the final source. Its source and menu bindings, and the independent exact scalar ledger, have passed. The analytic geometry and ledger have also received separate internal review. Numerical candidate searches are not part of the acceptance argument.

This extension keeps the matching types, outer reduction, trace aggregation, and all 62 previously certified branches. It adds one positive-containment lookup to three fixed circuit families. The resulting contact bounds are combined with the existing contacts by same-family chords and the finite contact-cone lemma.

III.1. Theorem and inherited proof interfaces

Keep the exact rational types

$$(a, b, s, t) = (.065, .075, .14, .45).$$

Theorem F3, restated. A uniform deterministic construction has a fixed startup constant $C_{\mathrm{start},3}$ and solves integer Subset Sum, including an indexed witness when one exists, in

$$C_{\mathrm{start},3} + \mathrm{poly}(n, L)\, 2^{E_3 n}, \qquad E_3 = \frac{158153814247997}{319021177919134} < .495748. \tag{T1}$$

Here L is the binary input length. The baseline cap by deterministic interleaving with the indexed meet-in-the-middle algorithm continues to apply. No new space bound or practical implementation of the compiled fast circuit is asserted.

The positive-containment factorization, signed inclusion-exclusion identity, first-lookup trace aggregation, integer matrix compiler, and output-sensitive incidence enumeration are the ingredients proved in Parts I–II. This extension changes the finite circuit family and the scalar gain. It does not replace any signed equality by a positive cover or infer a global proof from sampled input instances.

III.2. Exact third-lookup identity

For an outer color, use the notation of Parts I–II

$$\beta = a + \rho(m - a), \quad a \le m \le s, \quad m \le z \le \beta, \quad C_1 = \gamma_1 \beta, \quad Q_1 = \beta - z, \quad D_1 = C_1 - Q_1.$$

The first color satisfies

$$\max\{Q_1, C_1 - m\} \le w_1 \le \min\{\beta - m, C_1\}.$$

Retain the second-level routing

$$Q_2 = C_1 - w_1, \qquad C_2 = \max\{\gamma_2 C_1, Q_2\}, \qquad D_2 = C_2 - Q_2.$$

The branch $C_2 = Q_2$ is implemented by the existing first-level push circuit, with its previously proved upper profile. On the other closed branch put

$$v = w_1 - (1 - \gamma_2) C_1 \ge 0, \qquad C_2 = \gamma_2 C_1, \qquad D_2 = v.$$

Its second color obeys

$$\max\{Q_2, C_2 - Q_1\} \le w_2 \le \min\{D_1, C_2\}. \tag{T2}$$

After the second complement and transpose, the residual containment matrix has universe size C_2N, row size D_2N, and query size

$$Q_3N = (C_2 - w_2)N.$$

Indeed, (T2) implies $0 \le Q_3 \le D_2 \le C_2$. Choose a fixed rational $0 < \gamma_3 < 1$ and set

$$C_3 = \max\{\gamma_3 C_2, Q_3\}, \qquad D_3 = C_3 - Q_3. \tag{T3}$$

At integer sizes choose $\max\{\lceil \gamma_3 |C_2| \rceil, |Q_3|\}$. This contains the query and is at most $|C_2|$: for an integer $p \ge 0$, an integer $0 \le q \le p$, and $0 < \gamma < 1$, one has $\max\{\lceil \gamma p \rceil, q\} \le p$. There is no divisibility requirement on an inner lookup width.

The generic positive-containment factorization now gives the color domain and degree contributions

$$\max\{Q_3, C_3 - Q_2\} \le w_3 \le \min\{D_2, C_3\}, \tag{T4}$$

$$\begin{aligned} F_3 &= \max\{0, H(D_2, w_3) + H(Q_2, C_3 - w_3) - H(w_2, D_3)\}, \\ G_3 &= H(D_3, w_3 - Q_3). \end{aligned} \tag{T5}$$

Write L_0 for the outer occurrence exponent and F_2 for the second occurrence exponent. Let B_{tr} be the proved first-lookup trace bound, before adding the final residual degree. Three orientation swaps give

$$L^{(3)} = L_0 + F_2 + G_3, \qquad R^{(3)} = B_{\mathrm{tr}} + F_3. \tag{T6}$$

To check the orientation, the generic rule sends a residual pair (L', R') to $(F + \gamma R', \gamma L')$, with the scale converted to the original universe. Substitution of (T5) into the residual matrix after the second swap puts G_3 on the original row side and F_3 on the original query side. The earlier F_2 therefore stays on the row side.

The pooled signed coefficient is unchanged. After fixing the first outer and inner occurrence labels, every descendant incidence depends on the original completion U only through $K = U \cap C_1$. One first sums the completions with this trace, exactly as in the identity (47) in Part I, then applies the additional positive factorization to the residual matrix. The coefficient has $O(N)$ bits because it is a signed count of at most 2^N completions.

On the third push branch $Q_3 \ge \gamma_3 C_2$, (T3) gives $C_3 = Q_3$, $D_3 = 0$, and (T4) forces $w_3 = Q_3$. Thus

$$G_3 = 0, \qquad F_3 = H(D_2, Q_3) = H(D_2, w_2 - Q_2) = G_2. \tag{T7}$$

The last equality uses $Q_3 + (w_2 - Q_2) = D_2$ and entropy symmetry. This branch recovers the two-level recursive contribution exactly. The unmerged degree identities also agree on its routing boundary. The first-level push branch retains its older upper bound; that upper bound need not equal the sharper trace upper bound on their common boundary.

All routing domains and degree contributions are independent of the weight τ. The supremum of

$$\tau(L_0 + F_2 + G_3) + (1 - \tau)(B_{\mathrm{tr}} + F_3) \tag{T8}$$

over the closed third-level branches, together with the first-level push profile, is therefore convex in τ for each fixed family. Only contacts from that same family may be joined by a chord.

III.3. Families and full-domain hypotheses

The added menu is:

ρ	γ_1	γ_2	γ_3	Contacts (τ, U)
3.18	.728	.671	.583	(.44, .19449), (.48, .20164), (.50, .20530)
3.44	.721	.663	.584	(.48, .20187), (.50, .20507), (.52, .20834)
3.72	.714	.657	.589	(.52, .20809), (.56, .21380)

Every decimal is an exact rational. The maximum outer widths are .3035, .323, and .344, respectively. Each is below $\min\{t, 1 - t, 1 - s\}$. The first query must fit the first lookup on the entire outer domain, not merely on the boxes remaining after a filter. The affine function

$$\gamma_1 \beta(m) - (\beta(m) - m)$$

has the following positive values at its two endpoints:

ρ	At $m = a$	At $m = s$
3.18	.04732	.057448
3.44	.046865	.049883
3.72	.04641	.041616

Checking both endpoints proves the condition for every $m \in [a, s]$. All three inner fractions lie strictly between zero and one. The second and third max rules then contain every residual query, including degenerate zero-width cases. The final kernel rejects a family that fails any of these preliminary hypotheses.

The matching types are unchanged, so the original core-fitting, dummy-padding, and energy-table arguments retain their hypotheses. The scalar ledger nevertheless rechecks

$$s < 1/5, \quad s < 1/3, \quad 2s < 2/3, \qquad h(2s) + 2s < 1.135451 < 1.444238 < 1 + h(s) - s. \tag{T9}$$

The last comparison charges the initial difference table to the energy-transform budget; it does not require that table's exponent to be below one.

III.4. Exact four-dimensional acceptance

The final kernel uses box coordinates (m, z, v, θ) with

$$w_2 = Q_2 + \theta(D_1 - Q_2), \qquad 0 \le \theta \le 1. \tag{T10}$$

On every feasible first color, $Q_2 \le D_1$. Every legal second color therefore has a representation (T10). When $D_1 = Q_2$, all values of θ represent the same color. The initial box is

$$[a, s] \times [a, \beta(s)] \times [0, \gamma_2\gamma_1\beta(s)] \times [0, 1].$$

The first-color affine constraints and all four inequalities in (T2) are retained. The first-level push branch instead has $v = \theta = 0$ and uses its exact projected constraints and auxiliary-color elimination.

Coverage under tightening

Every expression affine in (m, z, v, w_2) becomes multiaffine after (T10): it is affine in each one of the four box coordinates while the other three are fixed. Its extrema over a rectangular box occur at corners.

For a multiaffine necessary constraint $g(x) \ge 0$, fix one coordinate $x_i = u$. The value of $g(u, x_{-i})$ is a convex combination of the affine functions $g_e(u)$ obtained by setting the other coordinates to their endpoints. Consequently a feasible u must satisfy $g_e(u) \ge 0$ for at least one such endpoint assignment. Intersect each exact affine half-line with the current coordinate interval, take the hull of their union, and intersect the box coordinate with that hull. This cannot remove a feasible point. An empty union or a strictly negative maximum of g proves the box empty.

This projection uses rational roots with known scalar coefficients. It never divides by an interval coefficient or assumes that a varying slope has one sign. Repeating it a fixed number of times changes only the search tree, not the set that must be covered.

Entropy bounds and auxiliary colors

For each rational $0 < p < 1$, the Gibbs plane

$$H(c, y) \le -y \log_2 p - (c - y) \log_2(1 - p) \tag{T11}$$

bounds every positive entropy on its domain. Floating arithmetic proposes p; the accepting plane uses exact rational logarithm enclosures.

For a negative entropy, test its entire box domain using all corners. The conditions $c \ge 0$, $y \ge 0$, and $c - y \ge 0$ are multiaffine, so their corner tests suffice. If they pass, the composition $H(c(x), y(x))$ is concave in each coordinate separately. Successive one-coordinate concavity proves that the multilinear interpolation of certified lower corner values lies below the entropy on the whole box. If any corner fails the domain test, use the valid lower bound zero on the actual feasible colors.

The four negative entropy pairs are $(1 - m, \beta - m)$, (z, D_1), (w_1, D_2), and (w_2, D_3). The last pair is used only for the third recursive branch.

After choosing Gibbs planes, the coefficient of the remaining scalar color w_3 is constant over a box. For a nonnegative coefficient, substitute any globally valid affine upper endpoint from (T4); for a negative coefficient, substitute a globally valid lower endpoint. The center may select

which endpoint is convenient, but the chosen expression is fixed throughout the box. Each choice is individually valid, so floating selection cannot invalidate the bound.

Retain the nonnegative floor in F_3 as two separate cases. For the raw case use (T4). For the zero case put $y = w_3 - Q_3$ and use

$$\max\{0, D_3 - Q_2\} \le y \le \min\{D_3, w_2 - Q_2\}. \tag{T12}$$

Eliminate y separately. The F_2 floor is likewise retained; the checker never assumes that either raw occurrence exponent is positive.

Routing and whole-box maxima

In the recursive expression, the third routing sign is

$$D_3 = w_2 - (1 - \gamma_3)C_2.$$

Its corner extrema certify a box lying wholly on either side. If a box crosses zero, take the maximum of the push and recursive majorants on the whole box. Their affine extensions need only bound their own feasible branch. Their maximum covers both branches.

After substitution, each plane and each interpolant is multiaffine. The floors and routing maxima preserve convexity in each coordinate separately. The resulting upper bound therefore attains its whole-box maximum at a corner. The old trace bound and continuous trace bound each receive a whole-box maximum before their minimum is taken. A pointwise minimum of corner values would not justify coverage and is not used.

Every accepted leaf has an exact rational upper bound at most its contact target. Empty leaves have a necessary-constraint proof. Binary subdivision covers the parent box, and the completed tree has exactly $2(\text{accepted} + \text{empty}) - 1$ nodes. Together these facts prove the continuous contact inequality. Node or depth limits stop with an error; they never certify an unresolved region.

Arithmetic and recovery

The logarithm enclosure normalizes its rational argument to $[1, 2]$ and uses the 32-term atanh series. With $q = (x - 1)/(x + 1)$, the omitted tail is bounded by $2q^{65}/(65(1 - q^2))$. Integer outward rounding at scale 2^{128} and exact rational arithmetic then enclose the logarithms used in (T11) and in the entropy lower bounds. The unchanged proof of this enclosure is the one used by the two-level checker.

The input-time Subset Sum algorithm and its exact uniform seed search do not call the floating proposal routine. That routine only helps find rational planes for this finite verification. Different numerical libraries may propose different planes or subdivision trees; every accepting comparison is still made by the same exact inequality. The only executable dependencies of the verifier are the Python standard library, its fixed menu, and the explicitly supplied mathematical proofs.

The optional recovery driver atomically preserves completed branches and exact JSON subdivision frontiers. It binds each record to the kernel, menu, and driver bytes. These records are trusted local work products, not independently verified proof objects. The ordinary clean replay starts from the initial boxes without loading them. Spawned workers explicitly check the same source hashes and load the fixed menu independently. An interrupted branch may therefore resume without relying on inherited mutable process state.

III.5. Uniform compiler allowance

The extra lookup introduces finitely many additional labels and one more integer ceiling. A constant number of sampling levels still uses $\exp(O(N))$ labelled events. The same polynomial oversampling and simultaneous bounds establish a finite exact circuit family. The first-lookup pooled histogram and the signed coefficient remain unchanged.

Choose integer widths with the max rules above. Their normalized values differ from the ideal widths by $O(1/N)$ at this fixed depth. Use the full pooled histogram expression (51) and the unmerged degree identities for this limit argument. Their entropy expressions are continuous on compact feasible domains, and every limit of rounded feasible colors belongs to one of the retained closed routing branches. Thus the three inner lookup ceilings, the unchanged outer rounding, polynomial color factors, and seed approximations incur a uniform $o(1)$ per-coordinate loss.

No continuity of a simplified minimum envelope is needed. Nor is continuity of the inverse quotient defining θ assumed at $D_1 = Q_2$. Rounding is carried out in the original raw color coordinates. Compactness supplies a legal limit, and (T10) represents that limit even when its interval collapses.

All old and new contact weights remain in $[.24, .76]$. The finite contact-cone amplification is therefore still at most $25/6$. Choose the common pre-cone seed loss below $3 \cdot 10^{-10}$, giving post-cone loss below $1.25 \cdot 10^{-9}$. Choose the common finite amplification depth so its further loss is below $1.25 \cdot 10^{-9}$. Their sum fits the strict template allowance $2.5 \cdot 10^{-9}$.

Keep the divisibility scale $H_0 = 1000$, microblocks $N_0 = k^2 + 1000k$, and core grid size 10000. Count descriptions and dummy padding contribute less than $2.5 \cdot 10^{-9}$. The exact interpolation loss before padding is $81/21875000000$; with sufficiently small fixed padding inflation it remains below $5 \cdot 10^{-9}$. These three allowances retain the total $\zeta = 10^{-8}$ used below.

The common fixed seed may have to increase. Exhaustive exact matrix checks still terminate because the finite circuit exists, and the larger search cost is absorbed in $C_{\mathrm{start},3}$. Input-time tensor products, stored adjacency lists, gate naming, and output-sensitive enumeration remain those proved in Parts I–II. The certificate's fourth coordinate creates no input-time continuous optimization.

III.6. Reflected envelope and complete outer ledger

Combine every old and new same-family chord, both endpoint circuits, and the two global cone pieces from each contact. For a contact (α, U) these pieces are

$$U\frac{1-\tau}{1-\alpha} \quad (0 \le \tau \le \alpha), \qquad U\frac{\tau}{\alpha} \quad (\alpha \le \tau \le 1).$$

The finite-volume proof applies to actual integer color degrees, so it also transfers the already rounded seed estimate. It introduces no new continuous contact premise.

For the stated menu the exact lower envelope has 18 breakpoints. The reflected quadratic bound requires 112 exact endpoint comparisons. Its maximum is

$$\frac{212647711284991364849543952290503}{518400000000000000000000000000000}, \tag{T13}$$

attained at $\tau = 49/100$ and $51/100$. Choose

$$S_3 = .41020007.$$

The exact margin is

$$S_3 - (\mathrm{T13}) = \frac{5003008635150456047709497}{518400000000000000000000000000000000} > 0. \tag{T14}$$

Retaining the same outward type constants, exact entropy enclosures give

$$g_3 \geq g_{3,L} = \frac{135677471157}{10^{13}}, \qquad \mu_3 = \frac{50000000000000}{159510588959567}. \tag{T15}$$

The gain includes the entire $\zeta = 10^{-8}$ compiler allowance. With $D_U = 58153814247997/10^{14}$,

$$\mu_3(1 + D_U) = \frac{1}{2} - \mu_3 g_{3,L} = E_3. \tag{T16}$$

The changed dictionary fraction slightly increases some nonbinding rates, so every operation is checked again:

Operation	Certified bound on the exponent
Circuit incidences and indexed witnesses	E_3
Canonical output and complete residue cover	E_3
Modulus energy search	$< .463275485$
Selectors and shared witness preprocessing	$< .321176660$
Repeated prefix lookups	$< .328894505$
Repeated outside lookups	$< .352437108$
Cyclic convolution and shift scans	$< .372294358$
High-deficiency fallback	$< .493730824$
Dictionary preparation	$< .313458814$

All seven nonbinding bounds are strictly below E_3. The exact ledger also verifies $3\mu_3 < 1$, $g_{3,L} < 1/50$, the canonical-layer density gate, all endpoint entropy enclosures, every envelope intersection and continuity junction, and the unchanged compiler and core-fitting requirements.

III.7. Evidence boundary

The final-source receipt contains all 16 new branch results, bound to the kernel and fixed menu. It records 317,138 accepted leaves, 634,268 nodes, four empty leaves and maximum depth 29. The smallest exact margin is

$$\frac{7341906319169228363258538380 1}{641663648000000000000000000000000000000} > 1.14 \cdot 10^{-8}.$$

The old 62-branch receipt remains separate and unchanged. The new scalar ledger uses both proved sets of continuous contact inequalities and the analytic construction; it does not prove these ingredients by itself. Its default command verifies Theorem F3, with 112 reflected-profile comparisons. The independent option `--base-two-level` reproduces all 160 comparisons and the preserved Theorem F2 exponent without reading the third-level menu.

Run the inexpensive binding and scalar checks from the repository root:

```
python3 tools/verify_subset_sum_third_contacts.py --receipt \
  problems/subset-sum-fast-deterministic/verify/third_receipt.json
python3 tools/verify_subset_sum_scalars.py
python3 tools/verify_subset_sum_scalars.py --base-two-level
python3 problems/subset-sum-fast-deterministic/verify/check_headline_ledger.py
```

Receipt binding checks provenance and complete result accounting; it does not rerun the box inequalities. The verification note gives the clean full replay and the optional resumable command, with their different evidence scopes.

An independent geometry audit checked feasible-color preservation, degenerate second-color intervals, multiaffine interpolation identities, mixed routing boxes, and the optimized closed trace guard. These finite checks accompany the analytic arguments above. They do not replace the full-domain branch replay or establish the complete Subset Sum theorem independently of the rest of this manuscript.

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